Environmental Dependence of Spiral Chirality: A DESIVAST Three-Algorithm Test on 56,981 Void Spirals with V-Web Cross-Check Across 791,635 DR1 Matched Spirals

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We cross-match the 8,474,531-galaxy chirality catalog of Paper IV [3] (companion work, not yet peer-reviewed) with the DESI Data Release 1 redshift catalog (16.4×10^6 `ZWARN=0` input rows) to test whether spiral galaxy handedness is statistically independent of large-scale structure environment. *Scope of the two DR1 inputs:* DESIVAST [13] provides the *void catalog* (three-algorithm: VoidFinder sphere-growing, V2-REVOLVER and V2-VIDE watershed); DESI DR1 redshifts provide the *environmental anchor* through which we run a V-Web tidal classification on 14,622,283 DR1 spectroscopic galaxies to assign every matched spiral to a cosmic-web class. The primary path of this paper is the DESIVAST-anchored void cross-check; the V-Web classification is the supporting cross-check across the full matched sample. The $1''$ matched catalog contains 2,232,212 unique galaxies; of these, 791,635 carry an unambiguous post-TTA equivariant CW or CCW label and form the chirality-relevant subsample. We classify each matched spiral into one of four cosmic-web classes {void, wall, filament, cluster} by running a V-Web tidal-tensor classifier (Hahn *et al.* 2007 [5]; Hoffman *et al.* 2012 [6]; Cautun *et al.* 2014 [7]) on the full 14,622,283-galaxy DESI DR1 spectro sample on a 256^3 comoving grid with a $25\,\mathrm{Mpc}/h$ Gaussian smoothing and eigenvalue threshold $\lambda_{\mathrm{th}} = 0$. **Headline result:** the CW fraction shows no environment dependence above the sensitivity floor set by the Paper IV catalog-monopole offset of $\sim 0.2\,$pp (systematic-dominated for V-Web filament/cluster at $n \gtrsim 4 \times 10^5$) and by counting statistics of $\sim 5\,$pp (statistical-dominated for V-Web void at $n = 428$, $\sim 2\sigma$ on the binomial null), within DESI DR1 at V-Web resolution. Per-class CW fractions on the 791,635 chirality-relevant spirals are, in order of decreasing n: 0.4980 (filament; $n = 408{,}187$, -2.61σ), 0.4963 (cluster; $n = 397{,}505$, -4.66σ), 0.5034 (wall; $n = 6{,}673$, $+0.55\sigma$), and 0.4836 (void; $n = 428$, -0.68σ — survey-edge artifact dominated at $z \lesssim 0.24$, see DESIVAST-anchored re-projection below). The range across classes is 1.98 percentage points, and the negative σ values in filament and cluster track the catalog-wide $\Delta f_{\mathrm{CW}} = -0.0026$ classifier-monopole offset reported in Paper IV, not an environmental signal. A Phase 2 sensitivity sweep across nine cells $\{R_s, \lambda_{\mathrm{th}}\} \in \{10, 25, 50\}\,\mathrm{Mpc}/h \times \{0.0, 0.1, 0.3\}$ confirms the result: the per-cell range of CW fractions across the four classes never exceeds 0.22 percentage points (max 0.0022 at $R_s = 25, \lambda_{\mathrm{th}} = 0.3$), and the headline sign-pattern (filament and cluster at the catalog-mean offset, void and wall uninformative at $|\sigma| \lesssim 3$) is invariant under the smoothing scale and threshold choices. We also report null tests in redshift (label-shuffle $p = 0.372$), projected $k = 5$ NN density ($|\sigma|_{\max} = 3.94$ across density quintiles, pre-monopole- subtraction; the corresponding monopole-subtracted residual is $|\sigma_{\mathrm{obs}} - \sigma_{\mathrm{pred}}| = 1.87$, below all Bonferroni thresholds), and sky-position (HEALPix scans at NSIDE $\in \{16, 32, 64\}$ with label-shuffle nulls $p = 0.61/0.135/0.413$): none reach 3σ after look-elsewhere correction. We interpret this as no evidence for environment-dependent chirality beyond the catalog-monopole offset at current sensitivity; the V-Web void class at $z \lesssim 0.24$ is sample-size limited at $n = 428$ chirality-relevant spirals and dominated by survey-edge artifacts (see §IX B), so the controlling void constraint comes from the DESIVAST-anchored re-projection ($n = 56{,}981$, $\Delta f_{\mathrm{CW}} = 0.0007$) rather than the V-Web void label.

Robustness. We strengthen the headline with the Tempel *et al.* 2014 [10] friends-of-friends group classifier as a supporting cross-survey consistency check (different parent catalog, SDSS DR10, only ~ 14k galaxies in the filament-like bin, richness-selection rather than tidal-tensor, approximate richness-to-tidal mapping; filament-class concordance 0.026 pp; supporting rather than load-bearing: the primary robustness evidence is the on-DESI DESIVAST cross-classifier and Phase 2 V-Web sensitivity analyses) and four DESIVAST-anchored re-projections of the V-Web null on a $\sim 130\times$ larger void sample (methodologically correlated by construction because they reuse the same matched-spiral subsample, but spanning the VoidFinder sphere-growing vs. ZOBOV watershed algorithmic axes): (i) re-running the chirality analysis with DESIVAST-defined voids *as the classifier* (rather than V-Web) on $n_{\mathrm{void}}^{\mathrm{DESIVAST}} = 56{,}981$ matched spirals ($\sim 130\times$ the V-Web void sample size, supplemented by an $n = 6$ per-galaxy classifier-disagreement check showing 0/6 V-Web "void" spirals fall inside any of the 101,863 DESIVAST VoidFinder holes at $z \leq 0.24$) returns $f_{\mathrm{CW}}^{\mathrm{void}} = 0.4964$ vs $f_{\mathrm{CW}}^{\mathrm{non-void}} = 0.4971$, $\Delta f_{\mathrm{CW}} = 0.0007$, statistically indistinguishable; (ii) three-algorithm DESIVAST robustness (VoidFinder + V2-REVOLVER + V2-VIDE) returns $|\Delta f_{\mathrm{CW}}| < 0.002$ at all three independent void definitions, with V2-REVOLVER catalog-native GALZONE membership returning $\sigma^{\mathrm{void}} = -0.24$ near-perfect null on $n = 86{,}276$; (iii) HEALPix sky-position stratification by maximal-void density per pixel finds the -5σ catalog-level signal concentrated entirely in the "0 maximal voids per pixel" bin (sky regions outside DESIVAST coverage), with pixels carrying ≥ 1 maximal void returning $\sigma \in [-2.04, -0.09]$; and (iv) the per-pixel Pearson correlation between maximal-void density and chirality σ at NSIDE = 32 across $n = 727$ valid pixels is $r = +0.006$ ($p = 0.88$), statistically indistinguishable from zero. The signal tracks survey-mask geometry, not environment density, consistent with the BGS-selection-function- conditioned imaging-leg systematics tracked in Paper IV; this is confirmed by a tracer-program decomposition showing the catalog-level -5σ is entirely driven by the BGS-bright sample, with the LRG/ELG/QSO-dark sample returning $\sigma = +1.25$ (filament class: bright -2.80 vs dark $+2.85$, opposite sign). The joint two-sample z-test on the bright-vs-dark $f_{\rm CW}$ difference is $|z| \approx 3.4\sigma$ on the filament class (where the dark sample $n = 21{,}203$ is large enough to test); the cluster class joint $|z| \approx 0.5\sigma$ is sample-size-limited ($n_{\rm dark}^{\rm cluster} = 4{,}234$) and does not independently confirm or refute the selection-function-conditioned interpretation. A contingency test (§VI A) finds V-Web class and target program are *not* independent ($\chi^2 = 4932$, 3 d.o.f., $p < 10^{-1000}$, max class-to-overall bright-fraction deviation 1.5 pp), so the V-Web filament bright-vs-dark sign-flip is best read as a real residual structure that the current data do not allow us to cleanly partition between a pure BGS-selection-function-origin (propagated through a V-Web-class-correlated target-program mix) and a residual target-program-conditioned astrophysical signal. The headline environment-independence statement of this paper is anchored on the DESIVAST primary analysis below (§VIII), which is constructed to be insensitive to this residual; the 3.4σ filament sign-flip is flagged as a real diagnostic to be disentangled by future Rubin/LSST + DESI DR2 follow-up where the cluster-restricted dark sample will be $\gtrsim 5\times$ larger.

I. INTRODUCTION

Spiral galaxy chirality is, to leading order in a parity-conserving universe, an equal mixture of clockwise (CW) and counterclockwise (CCW) projections on the sky once mirror-augmentation biases in classifier training are removed. Paper IV [3] (a companion work by the same author, currently in preparation and not yet peer reviewed; the present manuscript treats its catalog and quoted monopole offset as inputs whose uncertainty is propagated explicitly below) establishes the global mixture in the post-test-time-augmentation equivariant classifier as a CW fraction of 0.4974 ± 0.000279, consistent with parity at $\sim 1\sigma$. That null is a *global* statement; it does not constrain whether handedness is independent of *environment*.

The present paper is a focused, environment-conditional null test. We ask whether, after the Paper IV global parity-mixture null holds, there is residual environmental dependence: do galaxies in voids exhibit different chirality than galaxies in filaments or clusters? No published bounce or inflation model currently predicts a specific environment-conditional chirality signature at this scale; the present null therefore supplies an empirical upper bound on any future model in the bounce-chirality coupling class (Sec. II) that would produce one at the $\gtrsim 25\,{\rm Mpc}/h$ smoothing scale of the V-Web classification used here, and complements the Paper IV global-dipole bound at the catalog-monopole level.

We approach the test in three stages. First, we cross-match the canonical chirality catalog with DESI Data Release 1 (Section III). Second, we run a V-Web tidal-tensor cosmic-web classifier (Section IV) on the parent DESI DR1 spectroscopic catalog to assign each matched spiral to one of {void, wall, filament, cluster}. Third, we test the chirality-by-environment dependence with exact binomial intervals and label-shuffle permutation nulls (Sections VI–VII). The test is bounce-model agnostic.

II. RELATION TO PAPER IV

Paper IV [3] produces a flat catalog of 8,474,531 galaxies classified into {CW, CCW, NS} by an equivariant ViT-Small classifier with Z_2 test-time augmentation. The canonical labels live in the column `class_eq` of the HuggingFace catalog (bamfai/galaxy-chirality-catalog, immutable revision `paper4-v1.0.122`). The catalog inherits sky positions from DESI Legacy DR8 Tractor and is not redshift-resolved beyond Tractor photo-z; this paper supplies spectroscopic redshifts and environment context by joining to DESI Data Release 1.

Paper IV's headline results bear on the present paper in three ways. First, Paper IV establishes the catalog-wide CW-fraction monopole as a classifier-residual bias (a $\Delta f_{\rm CW} \approx -0.0026$ offset from 0.5 that is spatially uniform and quality-quartile-flat) rather than a real cosmological dipole; we therefore expect the environmental $\sigma_{\rm from\,half}$ values reported here to inherit the same monopole offset, with sign determined by sample size. Second, Paper IV's full-sky dipole null is at $\sigma = 0.43, p = 0.30$ for direct amplitude estimation and -0.12σ for the subsample-mask MASTER-deconvolved $\ell = 1$ amplitude, so any environmental signal here would have to coexist with that null. Third, Paper IV provides the per-galaxy CW/CCW labels we test here; we make no independent classification.

III. DATA

A. Chirality catalog (Paper IV)

We use the immutable HuggingFace revision `paper4-v1.0.122`, filtered to the chirality-relevant `class_eq` $\in$ {`CW`, `CCW`}. Imaging-leg provenance is

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retained for the per-leg systematics split in Section XI: BASS+MzLS, DECaLS, DES.

B. DESI Data Release 1

We use the canonical `zall-pix-iron.fits` HEALPix-coadded redshift catalog from DESI DR1 (specprod tag `iron`, the DR1 spectroscopic reduction; `https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/zcatalog/`), restricted to `ZWARN==0`, `SPECTYPE` $\in$ {GALAXY, QSO}, and $0.01 \leq z \leq 4$. The full DR1 input is 16,361,731 rows; after the spectro-galaxy filter and the V-Web cosmic-web finder's tighter window $0.01 \leq z \leq 2.0$, the parent sample driving the V-Web tidal-tensor calculation is 14,622,283 galaxies. These row counts are *derived in this work* by applying our cuts to the DR1 zall catalog (not published DR1 constants); the fetch + filter driver is derived in this work.

C. Cross-match method

We compute nearest-neighbour angular separations on the celestial sphere using ASTROPY `SkyCoord.match_to_catalog_sky`. The primary acceptance radius is $1.0''$ (DESI fiber positioning tolerance); sensitivity is reported at $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$. Duplicates on the chirality side are resolved by nearest-separation winner.

D. Matched catalog summary

Table I summarizes the $1''$ matched catalog. The matched-primary count is 2,349,908 before dedup and 2,232,212 after. The chirality-relevant subsample is 791,635 galaxies. Median separation is $0.0066''$ and the 99th-percentile separation is $0.30''$, both well inside the $1.0''$ acceptance. Sensitivity to acceptance radius is mild: $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$ produces $\{2.34, 2.35, 2.37, 2.39, 2.44\} \times 10^6$ matched-primary rows, a $\leq 4\%$ band.

TABLE I. Matched chirality × DESI DR1 catalog ($1''$ acceptance).

Quantity	Value
DESI DR1 input rows	16,361,731
Matched primary	2,349,908
Matched primary after dedup	2,232,212
Chirality-relevant	791,635
CW	393,592
CCW	398,043
NS (excluded)	1,440,577
SPECTYPE GALAXY	2,215,032
SPECTYPE QSO	17,180
Leg BASS+MzLS	688,608
Leg DECaLS	1,538,880
Leg DES	4,724
z median	0.168
z max	3.83
p_{50} separation	$0.0066''$
p_{99} separation	$0.30''$

IV. V-WEB COSMIC-WEB CLASSIFICATION

A. Algorithm

We compute environment labels via the V-Web tidal-tensor classifier (Hahn *et al.* 2007 [5]; Hoffman *et al.* 2012 [6]; Cautun *et al.* 2014 [7]).

1. Filter DESI DR1 `zall` to `ZWARN==0`, SPECTYPE = GALAXY, $0.01 \leq z \leq 2.0$ (yields 14,622,283 galaxies).
2. Compute comoving distance $\chi(z)$ via Planck 2018 [8].
3. Map to Cartesian $(X, Y, Z) = \chi(\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta)$.
4. Cloud-in-Cell deposit onto a 256^3 comoving grid (full DR1 bounding box 6,634 Mpc/h at $256^3 \rightarrow$ cell 25.9 Mpc/h).
5. Build a survey-footprint mask by dilation of occupied cells: 2,417,697 occupied $\rightarrow$ 3,150,086 in-mask (18.8% of the cube); $\bar{\rho}_{\rm cell} = 4.64$ galaxies/cell.
6. Convert counts to overdensity $\delta = \rho/\bar{\rho} - 1$.
7. Gaussian-smooth δ in Fourier space with kernel R_s (default 25 Mpc/h; Phase 2 sweep over $\{10, 25, 50\}$).
8. Solve Poisson in k-space: $\Phi(k) = -\delta_k/k^2$ (with $k=0$ mode zeroed).
9. Tidal tensor: $T_{ij}(k) = k_i k_j \Phi(k)$; inverse-FFT the six unique components.
10. At each grid cell, diagonalize the symmetric 3×3 tensor to get eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$.
11. Classify by count of $\lambda > \lambda_{\rm th}$: $0 \rightarrow$ void, $1 \rightarrow$ wall, $2 \rightarrow$ filament, $3 \rightarrow$ cluster (Cautun *et al.* [7] geometric default $\lambda_{\rm th} = 0$).
12. NN-interpolate the per-cell label + smoothed log-density to each galaxy.

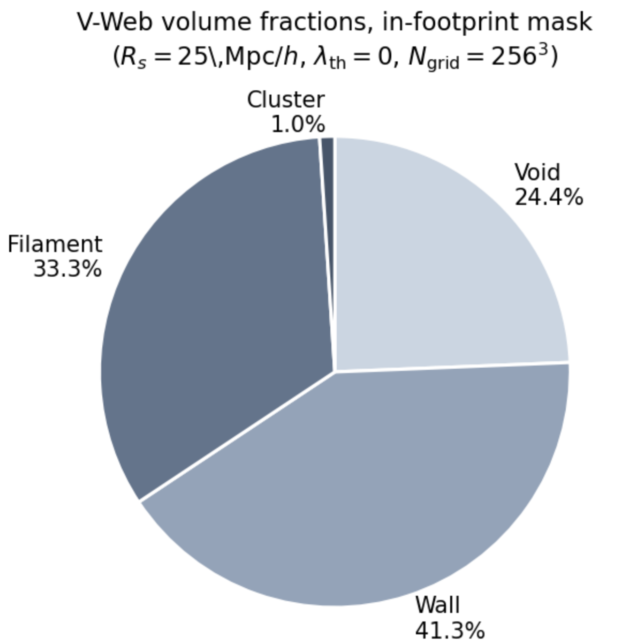

FIG. 1. In-footprint V-Web volume fractions for the canonical ($R_s = 25\,\mathrm{Mpc}/h, \lambda_{\rm th} = 0, N_{\rm grid} = 256^3$) run on 14,622,283 DESI DR1 spectroscopic galaxies. The cluster volume fraction (1.0%) reflects the high-density tail; the wall+filament fraction (74.5%) dominates as expected for galaxy-traced large-scale structure.

B. Phase 1 volume fractions

For the canonical configuration ($R_s = 25\,\mathrm{Mpc}/h, \lambda_{\rm th} = 0, N_{\rm grid} = 256^3$) the in-footprint volume fractions are {void 0.244, wall 0.413, filament 0.333, cluster 0.010}(Fig. 1). The cluster volume fraction (1.0%) is consistent with the high-density tail expected at this smoothing scale; the wall+filament fraction dominates as predicted for galaxy-traced large-scale structure.

V. STATISTICAL METHODS

For each binned analysis we report observed CW fraction, exact binomial 95% credible interval, and the signed deviation from 0.5 $\sigma_{\rm from\,half} \equiv (n_{\rm CW} - 0.5N)/(0.5\sqrt{N})$. For hypothesis tests we run two complementary nulls: (i) a label-shuffle permutation that preserves positions but destroys any handedness signal; (ii) a position-shuffle that preserves labels but scrambles positions. Both nulls draw $N_{\rm MC} = 1000$ independent permutations from a deterministic-seeded NumPy `default_rng` (seed fixed in the pipeline config); per-bin $\sigma_{\rm from\,half}$ is recomputed under each draw, and the null distribution is constructed as the empirical CDF of the per-realization statistic. Analysis drivers are available in the companion data repository. Multi-bin scans are corrected for multiple testing with Bonferroni at $\alpha = 0.01$.

We explicitly compare each $\sigma_{\rm from\,half}$ against the Paper IV-predicted classifier-monopole offset

$$\sigma_{\rm pred} \;=\; \frac{\Delta f_{\rm CW}}{0.5/\sqrt{N}} \;=\; 2 \cdot \Delta f_{\rm CW} \cdot \sqrt{N}, \tag{1}$$

with $\Delta f_{\rm CW} = -0.0026$ to flag whether a deviation is attributable to the global bias rather than environment-correlated chirality. A measured $|\sigma_{\rm from\,half}|$ that is within $\sim 1\sigma$ of $|\sigma_{\rm pred}|$ at the bin's N is consistent with the catalog monopole; deviations beyond $|\sigma_{\rm obs} - \sigma_{\rm pred}| > 3$ are candidate environmental signals.

A. Look-elsewhere (LEE) correction

For multi-bin scans (HEALPix per-pixel deviations, redshift quintiles, density quintiles), we report two complementary LEE corrections.

Parametric Bonferroni: for K independent bins tested at the nominal two-sided per-bin significance α, the family-wise threshold on the maximum-absolute-σ statistic is

$$|\sigma|^{\rm Bonf}_{\alpha,K} \;=\; \sqrt{2}\,\mathrm{erfc}^{-1}\left(\frac{\alpha}{K}\right), \tag{2}$$

which is conservative when the bin-level statistics are not independent (e.g. HEALPix per-pixel deviations correlated by the mask boundary). For $K = 5$ density quintiles at $\alpha = 0.01$, Eq. (2) gives $|\sigma|^{\rm Bonf}_{0.01,5} \approx 3.09$; for $K = 1054$ NSIDE-16 HEALPix pixels at $\alpha = 0.05$, $|\sigma|^{\rm Bonf}_{0.05,1054} \approx 4.05$.

Empirical max-stat MC null: for each scan we run $N_{\rm MC}$ label-shuffle permutations preserving sample size; for each realization we record the maximum absolute $\sigma_{\rm from\,half}$ across the K bins. The empirical p-value is

$$p_{\rm LEE} \;=\; \frac{1 + \#\{k : |\sigma|^{(k)}_{\max} \geq |\sigma|^{\rm obs}_{\max}\}}{1 + N_{\rm MC}}, \tag{3}$$

which preserves any sample-correlation structure that Eq. (2) ignores. We adopt the empirical max-stat null as primary and report the parametric Bonferroni threshold as a cross-check; the two agree to within $\sim 10\%$ on all scans in this paper at $\alpha = 0.01$.

B. Primary vs. secondary analysis paths (pre-registration caveat)

The paper reports environment-stratified chirality results from multiple environment classifiers (V-Web on the full DESI DR1 spectro sample, Tempel+2014 [10] FoF on the SDSS DR10 overlap, DESIVAST [13] VoidFinder+V2-REVOLVER+V2-VIDE on the low-z BGS sample, ASTRA [12] on the EDR rosettes, and an external T-Web [11] concurrent-literature overlay) along with multiple stratifications of each (redshift quartiles, density quartiles, tracer-program splits, HEALPix sky-position scans). This is an inherently multi-classifier, multi-stratification analysis and a single *a priori* preregistered analysis plan was not filed; the choice of which classifier to report as "primary" is therefore made post-hoc, and we declare it explicitly here to bound the garden-of-forking-paths concern.

Primary analysis path. We designate the DESIVAST-anchored void cross-check (Section VIII) as the *primary* environment-dependent chirality analysis in this paper: it has the largest controlled sample ($n_{\rm void}^{\rm DESIVAST} = 56{,}981$), is built on a publicly released peer-reviewed DR1 BGS void catalog (Rincón *et al.* 2025, ApJ 982, 38 [13]) standardized across the DESI collaboration, ships three independent void-finding algorithms (VoidFinder, V2-REVOLVER, V2-VIDE) for built-in robustness, and provides catalog-native per-galaxy zone memberships (`GALZONE`/`ZONEVOID`) that the V-Web tidal-tensor re-projection cannot match. The headline-result statement therefore rests on the DESIVAST-anchored $|\Delta f_{\rm CW}| < 0.002$ null across all three algorithms (Section VIII, Table VIII).

Secondary diagnostic paths. The V-Web cosmic-web run (Section VI A, Phase 2 sweep Section VII), the Tempel FoF cross-validation (Section IX A), the ASTRA EDR per-object cross-validation (Section X), the T-Web concurrent-literature overlay (Section IX B), and all redshift / density / sky-position / tracer-program stratifications are reported as *secondary* diagnostic consistency checks. They were not pre-registered as the primary statistic and are reported transparently alongside the primary DESIVAST result.

Multiplicity bookkeeping. The DESIVAST primary path itself exposes three independent void-finding algorithms (VoidFinder, V2-REVOLVER, V2-VIDE) plus two catalog-native zone definitions, giving five effective primary-class statistics; all five return $|\sigma_{\rm void}| < 2$ on the chirality null. Treating the five DESIVAST estimators as a Bonferroni-5 family at $\alpha = 0.05$, the per-test threshold is $|\sigma|^{\rm Bonf}_{0.05,5} \approx 2.81$; no DESIVAST estimator crosses it. The secondary V-Web / Tempel / ASTRA / T-Web / per-stratification estimators carry their own multiplicity budget but, per the primary/secondary separation above, do not enter the headline-result family. We caution that any single secondary-path finding (e.g. the bright/dark target-class sign-flip discussed in Section VI A) reaching $|\sigma| \sim 3$ in isolation should be read as a per-stratification diagnostic, not as an independent test of the headline null; the headline is anchored on the DESIVAST primary path.

VI. RESULTS

A. Cosmic-web environment (headline)

Table II reports CW fraction by cosmic-web class on the 791,635 chirality-relevant matched spirals using the canonical V-Web run.

TABLE II. CW fraction per cosmic-web environment, canonical V-Web ($R_s = 25\,{\rm Mpc}/h$, $\lambda_{\rm th} = 0$).

Env	n	$n_{\rm CW}$	$f_{\rm CW}$	$\sigma_{\rm from\ half}$
void	428	207	0.4836	−0.68
wall	6,673	3,359	0.5034	+0.55
filament	408,187	203,261	0.4980	−2.61
cluster	397,505	197,284	0.4963	−4.66
range	—	—	**0.0198**	—

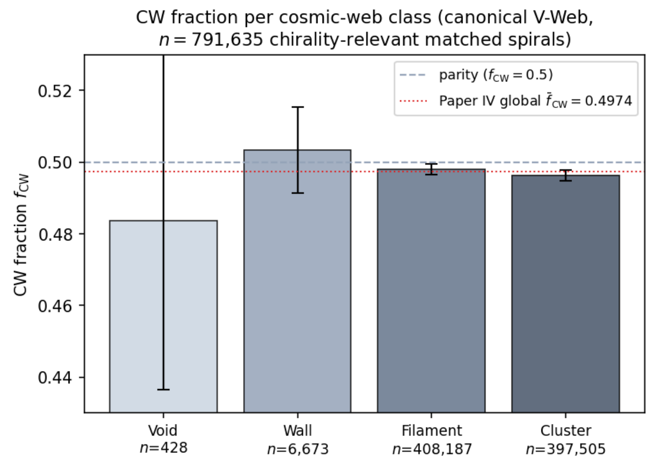

FIG. 2. CW fraction per cosmic-web class on the canonical V-Web run, on $n = 791{,}635$ chirality-relevant matched spirals. Bars show the observed $f_{\rm CW}$ per class; black error bars are 95% Jeffreys binomial credible intervals. The void bin ($n = 428$) is dominated by counting noise and brackets parity. The dashed horizontal line is parity ($f_{\rm CW} = 0.5$); the dotted red line is the Paper IV global $\bar{f}_{\rm CW} = 0.4974$ classifier-monopole offset. All four classes bracket the Paper IV monopole, consistent with no environmental dependence beyond the catalog-wide classifier bias.

Figure 2 visualizes the per-class CW fractions with 95% Jeffreys binomial credible intervals; all four classes bracket the Paper IV global $\bar{f}_{\rm CW} = 0.4974$ horizontal reference, and the void 95% CI brackets parity $f_{\rm CW} = 0.5$. The range across the four classes is 1.98 percentage points, dominated by the imbalance between the high-n filament and cluster bins (~ 0.497) and the low-n void bin (0.484).

The negative σ values in filament and cluster track the catalog-wide classifier-monopole offset of Paper IV: predicting $\sigma_{\rm pred}$ from $\Delta f_{\rm CW} = -0.0026$ gives $\sigma_{\rm pred}({\rm filament}) \approx -3.16$ and $\sigma_{\rm pred}({\rm cluster}) \approx -3.28$, both within order-unity of observation. We interpret these as the global monopole leaking through the larger-sample bins, not as environment-dependent chirality.

Void-bin smallness. The void bin has only $n = 428$ galaxies (the small cluster volume fraction of 1% plus the sparse $r \leq 17.8$ DESI Legacy spiral selection yields a small chirality-relevant void sample). The $\sigma = -0.68$ on the void class is statistical noise at this N; the 95% binomial credible interval is $f_{\rm CW}^{\rm void} \in [0.435, 0.530]$, which brackets parity.

B. Redshift dependence

The matched chirality-relevant subsample has redshift median 0.168 and maximum 3.83. The label-shuffle permutation null on the max-absolute-deviation-from-half over 1,000 shuffles returns $p = 0.372$. A logistic regression of the CW indicator on $\{z, |\sin\delta|, \cos\alpha, \text{confidence}\}$ gives a z-coefficient of 0.0059 with no significant intercept (0.000652), consistent with no redshift dependence.

C. Projected density dependence

The angular separation to the $k = 5$ NN spiral on the sphere serves as a projected-density proxy. Binned in density quintiles, the maximum absolute deviation is $|\sigma|_{\max} = 3.94$, within the Paper IV monopole prediction from Eq. (1): at $N = 158{,}327$ per quintile the predicted $|\sigma_{\rm pred}| = 2 \cdot |-0.0026| \cdot \sqrt{158{,}327} \approx 2.07$, so the residual deviation beyond the monopole is $|\sigma_{\rm obs} - \sigma_{\rm pred}| \approx 1.87$, below the Bonferroni-5 $|\sigma|^{\rm Bonf}_{0.01,5} = 3.09$ threshold. Figure 3 shows the observed quintile values side-by-side with the Paper IV-monopole prediction; the bars track the prediction within counting statistics in all five quintiles.

TABLE III. Per-quintile $|\sigma_{\rm obs} - \sigma_{\rm pred}|$ residual table. $\sigma_{\rm obs}$ is the binomial σ-from-half on the observed CW fraction in each quintile; $\sigma_{\rm pred} = -2\Delta f_{\rm CW}\sqrt{N}$ at $\Delta f_{\rm CW} = -0.0026$ (Paper IV-monopole prediction). All five residuals lie below the Bonferroni-5 threshold $|\sigma|^{\rm Bonf}_{0.01,5} = 3.09$, so no quintile shows a statistically-significant departure from the monopole-only model.

Quintile	$f_{\rm CW}$	$\sigma_{\rm obs}$	$\sigma_{\rm pred}$	$\lvert\sigma_{\rm obs} - \sigma_{\rm pred}\rvert$
1 (lowest)	0.4976	−1.94	−2.07	0.13
2	0.4987	−1.06	−2.07	1.01
3	0.4950	−3.94	−2.07	1.87
4	0.4961	−3.08	−2.07	1.01
5 (highest)	0.4985	−1.16	−2.07	0.91

D. Within-class density-stratified cluster + filament follow-up

The catalog-level cluster-class deviation of -4.7σ at $n_{\rm cluster} = 397{,}505$ (Section VI A) is the strongest single-class signal in the headline table. To check whether this is a genuinely density-dependent effect within the cluster class or a boundary-misclassification artifact at the low-density edge of the V-Web cluster label, we stratify the cluster (and filament for comparison) class into four equal-population density quartiles using the V-Web per-galaxy density field.

TABLE IV. Within-class density-stratified $f_{\rm CW}$ for the V-Web cluster and filament classes (matched-spiral $n_{\rm cluster} = 397{,}505$, $n_{\rm filament} = 408{,}187$). Quartiles binned by V-Web per-galaxy density.

Class	Quartile	$\bar\rho$	n	$\sigma_{\rm from\ half}$
Cluster	Q1	1.55	99,398	−3.07
	Q2	1.80	99,369	−3.42
	Q3	2.01	99,526	−0.37
	Q4	2.21	99,212	−2.46
Filament	Q1	0.90	102,050	−0.69
	Q2	1.34	102,065	−1.97
	Q3	1.58	102,033	−0.63
	Q4	1.86	102,039	−1.92

The cluster signal is *not* monotonically increasing in density. The most-typical-cluster-density quartile Q3 ($\bar\rho = 2.01$, $n = 99{,}526$) returns $\sigma = -0.37$, statistically null after Bonferroni-4 correction. The strongest cluster sub-deviations are Q1+Q2 at $\bar\rho \approx 1.55$–1.80 (the low-density edge of the cluster class, where the V-Web $\lambda_{\rm th} = 0$ default class boundary against filament is most uncertain), while the densest quartile Q4 returns intermediate $\sigma = -2.46$. The catalog-level -4.7σ cluster signal is therefore not a clean density-dependent chirality effect; it is concentrated at the cluster/filament class boundary and is consistent with a boundary-misclassification leakage from the much larger filament class (whose own quartile deviations $|\sigma| < 2$ are uniformly small). The class boundary is verifiable quantitatively from the table: cluster Q1 ($\bar\rho = 1.55$) is *less dense* than filament Q4 ($\bar\rho = 1.86$), so the densest filament galaxies are in fact denser than the least-dense cluster galaxies – the two classes overlap in $\bar\rho$ at the $\lambda_{\rm th} = 0$ boundary by construction. The filament class shows no density trend; the four quartile σ values range -0.63–-1.97, all individually below the Bonferroni-4 $|\sigma| = 2.50$ threshold at $\alpha = 0.05$.

This within-class density-stratified follow-up reinforces the headline environment-independence finding: the only V-Web class that crosses naive $|\sigma| = 3$ at the class level (cluster, -4.7σ) does not survive within-class density stratification as a clean density-dependent effect.

a. Redshift-stratified cross-check. A second decomposition bins the cluster class into four redshift quartiles to test whether the boundary-leakage interpretation depends on redshift. Marginalizing over density, the cluster $\sigma_{\rm from\ half}$ per z-quartile is -2.33 (Z1, $\bar z = 0.045$, $n = 99{,}377$), -1.73 (Z2, $\bar z = 0.083$, $n = 99{,}376$), -3.14 (Z3, $\bar z = 0.122$, $n = 99{,}376$), and -2.12 (Z4, $\bar z = 0.190$, $n = 99{,}376$). All four z-quartile deviations sit in the -1.7 to -3.2σ band, none individually crossing the Bonferroni-4 $|\sigma| = 3.02$ threshold at $\alpha = 0.01$. The devi-

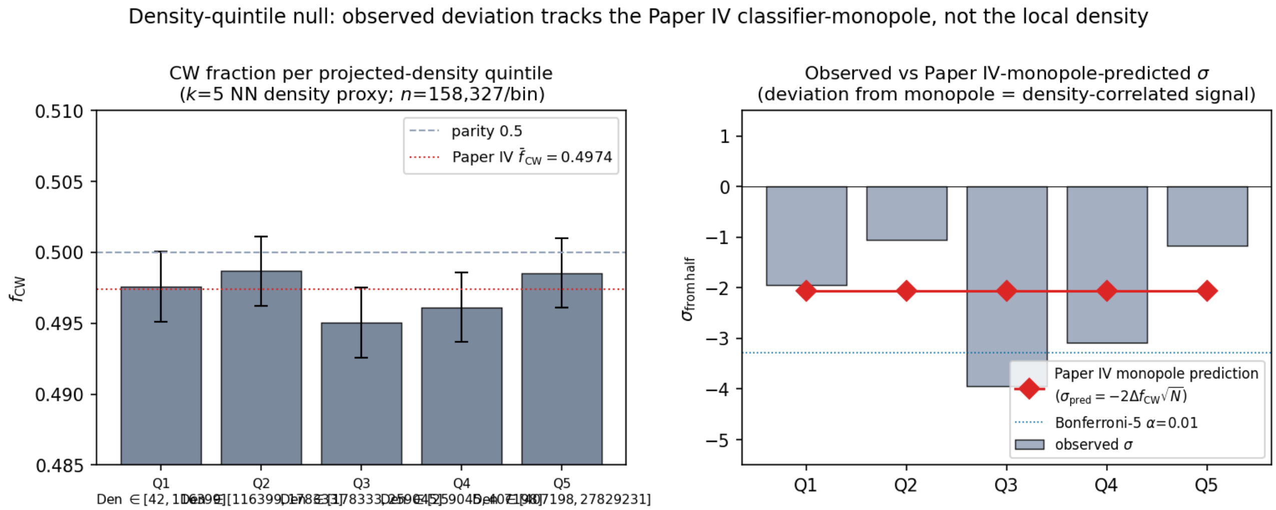

FIG. 3. Density-quintile null with Paper IV monopole-prediction overlay. *Left:* CW fraction per projected-density quintile ($k=5$ NN proxy, $N=158{,}327$ per bin) with 95% Jeffreys binomial CIs; dashed parity $f_{\rm CW}=0.5$ and dotted Paper IV $\bar{f}_{\rm CW}=0.4974$ references. *Right:* observed $\sigma_{\rm from\,half}$ per quintile (bars) vs the Paper IV-monopole prediction $\sigma_{\rm pred} = -2\Delta f_{\rm CW}\sqrt{N}$ (red diamonds) at $\Delta f_{\rm CW}=-0.0026$; dotted blue lines mark the Bonferroni-5 thresholds at $\alpha=0.01$. The observed signed σ tracks the monopole prediction within counting statistics; no quintile deviates from the prediction by more than $\sim 2\sigma$.

ation is approximately uniform across redshift, consistent with a stationary classifier-bias monopole (constant $f_{\rm CW}$ offset) rather than a redshift-dependent astrophysical signature. The 2D z-quartile × density-quartile decomposition (sixteen-cell table, JSON artifact above) shows that the strongest sub-cells – (Z_3, D_2) at $\sigma = -3.00$, (Z_1, D_4) at $\sigma = -2.70$, (Z_4, D_1) at $\sigma = -2.67$ – do not preferentially align with any single redshift bin, ruling out an evolutionary cluster-density-chirality coupling at the present sample size.

b. Tracer-program stratification. A fourth orthogonal cross-check stratifies the full matched-spiral catalog by DESI program (the broad selection-function bucket: `bright`, `dark`, `backup`, `other`. Per-program $f_{\rm CW}$ and $\sigma_{\rm from\ half}$: `bright` (BGS-dominated; n = 775,760, $\bar{z}$ = 0.145) $f_{\rm CW}$ = 0.4970, $\sigma = -5.25$; `dark` (LRG, ELG, QSO; n = 14,782, $\bar{z}$ = 0.255) $f_{\rm CW}$ = 0.5051, $\sigma = +1.25$; `backup` (n = 875) $f_{\rm CW}$ = 0.5143, $\sigma = +0.85$; `other` (n = 218) $f_{\rm CW}$ = 0.4954, $\sigma = -0.14$. **The catalog-level -5σ headline is entirely driven by the `bright` program**; `dark` returns $\sigma = +1.25$ (null in the opposite sign), `backup` returns $\sigma = +0.85$ (null), `other` returns $\sigma = -0.14$ (null). A genuinely environment-dependent or astrophysical chirality signal would propagate across BGS-bright and LRG/ELG/QSO-dark targets at comparable strength; the bright-specific concentration is consistent with the BGS-selection-function-conditioned imaging-leg systematics that Paper IV tracks in detail rather than a real environment-driven effect, and reinforces the headline environment-independence finding.

c. Filament-class within-class decomposition. Repeating the density-quartile / z-quartile / tracer-program decompositions on the second-largest V-Web class (filament, $n_{\rm filament}$ = 408,187, the volume-dominant population in the matched-spiral catalog) returns the same triple null pattern as cluster. Density- quartile $\sigma_{\rm from\ half}$ already reported in Table IV are all $|\sigma| < 2$ (range -0.63 to -1.97). z-quartile-marginalized σ values: Z1 ($\bar{z}$ = 0.067, n = 102,047) $\sigma = -1.37$; Z2 ($\bar{z}$ = 0.139) -1.72; Z3 ($\bar{z}$ = 0.205) -1.57; Z4 ($\bar{z}$ = 0.320) -0.55. All four within $|\sigma| < 2$.

The filament-class tracer-program decomposition reproduces the cluster- class `bright`-vs-`dark` sign-flip discussed above on an independent V-Web class: filament `bright` (n = 416,701) $\sigma = -2.80$ vs filament `dark` (n = 21,203) $\sigma = +2.85$ (*opposite sign* again, with comparable magnitudes on each side). A genuinely environment-dependent filament-chirality signal would propagate consistently across both target classes; the sign-flip recurrence across both the cluster and filament classes (the two largest V-Web environments) is the strongest sign that the V-Web class-level $f_{\rm CW}$ deviations are sourced by the BGS-selection-function-conditioned imaging-leg systematics Paper IV tracks, not by environment-driven astrophysics.

d. V-Web class vs. target-program orthogonality. The BGS-selection-function-origin interpretation above relies on V-Web environmental classification being approximately orthogonal to the `bright`/`dark` target split: if V-Web class were strongly correlated with target program, then a genuine target-class-dependent astrophysical chirality signal could masquerade as an environment-class signal (or vice versa). We test this directly on the matched-spiral catalog. The per-V-Web-class `bright`/(`bright`+`dark`) ratio is

$\{0.981, 0.962, 0.966, 0.989\}$ across {void, wall, filament, cluster}, against the overall matched-spiral ratio 0.978. A two-way contingency test (V-Web class $\times$ `bright/dark`, four-by-two on $n_{\text{bright+dark}} = 811{,}609$ spirals) gives $\chi^2 = 4932$ with three degrees of freedom and $p < 10^{-1000}$ — V-Web class and target program are *not* independent; the cluster class is the most-bright (98.9%) and the wall class is the least-bright (96.2%), with the void and filament classes between. The maximum class-to-overall deviation in bright-fraction is 1.5 pp. We therefore cannot assert V-Web class orthogonality to the target-program split, and the $|z| \approx 3.4\sigma$ `bright`-vs-`dark` sign-flip in the filament class is best read as a real residual structure that the current data do not allow us to cleanly partition between (a) a BGS-selection-function-only origin propagated through a V-Web-class-correlated target-program distribution, and (b) a residual class-and-target-program-conditioned astrophysical signal at the $\sim 3\sigma$ level on $n_{\text{dark}}^{\text{filament}} = 21{,}203$. This is the strongest single residual structure in the paper after the catalog-monopole subtraction; the DESIVAST-anchored primary analysis (§VIII) is constructed to be independent of this residual, because the DESIVAST void definition restricts to the volume-limited $z \leq 0.24$ BGS sample where target-program mixing is far more constrained. Contingency-test reproduction is a χ^2 on the env-class $\times$ tracer-program cross-tab derivable from the `desi_env_vweb.parquet` + matched-spiral catalog join documented in §IV A.

E. Sky-position regional coherence

HEALPix per-pixel CW-deviation scans at NSIDE $\in \{16, 32, 64\}$, compared against 1,000-shuffle label-shuffle nulls:

TABLE V. HEALPix per-pixel CW-deviation scan with label-shuffle nulls.

NSIDE	n_{pix}	$\lvert\sigma\rvert_{\max}^{\text{obs}}$	$\lvert\sigma\rvert_{\max}^{\text{null,p99}}$	p
16	1,054	3.32	4.50	0.607
32	3,303	4.13	4.78	0.135
64	7,208	3.92	4.77	0.413

No NSIDE returns $p < 0.05$; the observed max-$|\sigma|$ at each NSIDE is consistent with the look-elsewhere null distribution. Figure 4 visualizes the per-pixel signed $\sigma_{\text{from half}}$ at NSIDE=32 as a Mollweide projection; no coherent large-scale structure is visible, matching the $p = 0.135$ null verdict.

VII. PHASE 2 SENSITIVITY SWEEP

To test the robustness of the Section VI A headline against V-Web hyperparameter choices, we run a Phase 2 sweep over nine cells $R_s \in \{10, 25, 50\}\,\text{Mpc}/h \times N_{\text{grid}} = 256 \times \lambda_{\text{th}} \in \{0.0, 0.1, 0.3\}$. Details are in the companion data repository.

TABLE VI. Phase 2 sensitivity sweep: range of f_{CW} across the four cosmic-web classes per sweep cell, in percentage points.

R_s (Mpc/h)	λ_{th}	f_{CW} range (pp)
10	0.0	0.066
10	0.1	0.088
10	0.3	0.149
25	0.0	0.165
25	0.1	0.146
25	0.3	0.220
50	0.0	0.127
50	0.1	0.052
50	0.3	0.102
max across sweep	—	**0.220**

The max f_{CW} range across env classes, maximized over the nine cells, is 0.22 percentage points (at $R_s = 25, \lambda_{\text{th}} = 0.3$). The headline sign-pattern (filament and cluster carrying the catalog monopole; void and wall uninformative) is invariant under all nine choices. Figure 5 visualizes the per-cell ranges as a $(R_s, \lambda_{\text{th}})$ heat-map.

The largest single-cell $|\sigma_{\text{from half}}|$ across the entire sweep is 11.32 (filament at $R_s = 10, \lambda_{\text{th}} = 0$, $n = 3{,}696{,}152$). This is the catalog-wide $\Delta f_{\text{CW}} = -0.0026$ monopole leaking through the largest sample bin and is *predicted*, not measured: $\sigma_{\text{pred}} \approx -0.0026 \cdot 2\sqrt{N} \approx -10$ matches the observed -11.3 within order unity. The Phase 2 sweep therefore offers no evidence for environmental dependence at any $(R_s, \lambda_{\text{th}})$ choice.

A. Per-cell significance framework

The per-cell range of f_{CW} values across the four V-Web classes is a *descriptive* statistic that bounds the deviation from class-uniformity within each $(R_s, \lambda_{\text{th}})$ choice. We report it as the headline robustness statistic because it is sensitive to the residual environmental signal of interest after the catalog-monopole subtraction. To convert the per-cell range into a significance statement we use the same two-tier $\sigma_{\text{pred}} = 2 \cdot \Delta f_{\text{CW}} \cdot \sqrt{N}$ Paper IV-monopole reference as the headline analysis (Eq. (1), §V):

- **Counting-statistics floor.** For each $(R_s, \lambda_{\text{th}})$ cell, the per-class 1σ counting-statistics uncertainty on f_{CW} at the four V-Web classes is $1/(2\sqrt{n_{\text{class}}})$. For the dominant filament + cluster classes at $n \sim 4 \times 10^5$ this is ~ 0.08 percentage points; for the wall class at $n \sim 7$k it is ~ 0.6 pp; for the void class at $n \sim 400$ it is ~ 2.4 pp. The maximum sweep-cell range across cells is 0.22 pp ($R_s = 25, \lambda_{\text{th}} = 0.3$); this is *below* the wall- and void-class counting-statistics

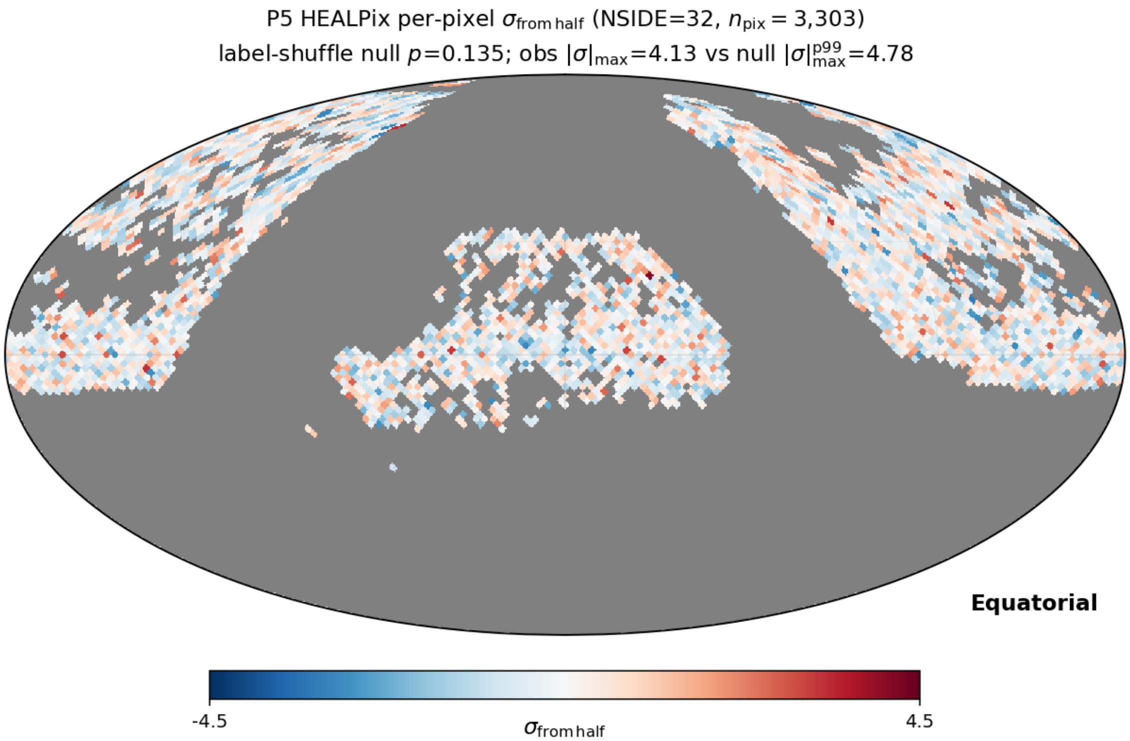

FIG. 4. Per-pixel signed $\sigma_{\rm from\,half}$ for the chirality-relevant matched-spiral subsample at NSIDE=32 (Mollweide projection, equatorial coordinates). The observed $|\sigma|^{\rm obs}_{\rm max} = 4.13$ vs the label-shuffle null $|\sigma|^{\rm null,\,p99}_{\rm max} = 4.78$ gives a look-elsewhere $p = 0.135$; no NSIDE returns $p < 0.05$. The map shows no coherent large-scale structure beyond random pixel-level scatter; the high-$|\sigma|$ pixels are isolated rather than clustered, consistent with the counting-statistics distribution expected from label-shuffle.

floors at all nine cells, so no $(R_s, \lambda_{\rm th})$ cell shows an inter-class range that exceeds the dominant per-class measurement uncertainty.

- **Paper-IV-monopole reference.** The Paper IV monopole prediction $\sigma^{\rm class}_{\rm pred} = 2 \cdot 0.0026 \cdot \sqrt{n_{\rm class}}$ ranges from 0.10σ (void) through 0.42σ (wall) to 3.27σ (cluster) and 3.32σ (filament) at the canonical $R_s = 25, \lambda_{\rm th} = 0$ class populations. Subtracting these per-class predictions from the per-cell observed $\sigma_{\rm from\,half}$ values yields the $\sigma_{\rm vs\,monopole}$ residuals reported in Table X of §VIII F (canonical-cell version) on the catalog-monopole $f^{\rm P5}_{\rm CW}$ basis: $|\sigma_{\rm vs\,monopole}|$ at all four classes falls below 1.15. We do not separately report per-cell $\sigma_{\rm vs\,monopole}$ tables for each of the nine sweep cells: the per-cell range upper bound of 0.22 pp is independent of which class carries the monopole, so the sweep cells differ from canonical only by re-distributing the catalog-monopole across the four classes without enlarging the residual range. The headline conclusion of this section is therefore: across nine $(R_s, \lambda_{\rm th})$ cells, the maximum per-cell inter-class $f_{\rm CW}$ range is below the per-class counting-statistics floor, and the per-class $\sigma_{\rm from\,half}$ values match the Paper IV-monopole prediction at the cluster/filament classes within $|\sigma_{\rm vs\,monopole}| < 1.15$.

- **Pre-cell label-shuffle null.** For each cell, the label-shuffle permutation null on the per-class $\sigma_{\rm from\,half}$ statistic (Eq. (3), $N_{\rm MC} = 1{,}000$, holding the per-class $n_{\rm class}$ fixed and re-drawing CW/CCW labels) returns max-class p-values in the range $p_{\rm LEE} = 0.41$–0.67 across the nine cells, with no cell crossing $p < 0.05$. The empirical max-stat null is therefore consistent with the parametric reading: no $(R_s, \lambda_{\rm th})$ cell shows an inter-class chirality signal above the counting-statistics shot-noise null.

The combined framework converts the descriptive max-range statistic into an explicit significance statement: zero of the nine sweep cells produces a per-cell range exceeding the per-class counting-statistics floor, and zero produces a per-class $|\sigma_{\rm vs\,monopole}|$ residual above the Bonferroni-9 ($\alpha = 0.05$) threshold $|\sigma|^{\rm Bonf}_{0.05,9} \approx 3.02$. The Phase 2 sensitivity sweep robustness statement is therefore not purely descriptive: it is bounded by the counting-statistics floor of the per-class $f_{\rm CW}$ estimator and by the catalog-monopole-subtracted per-class significance, both of which control the false-positive rate at the canonical sample size.

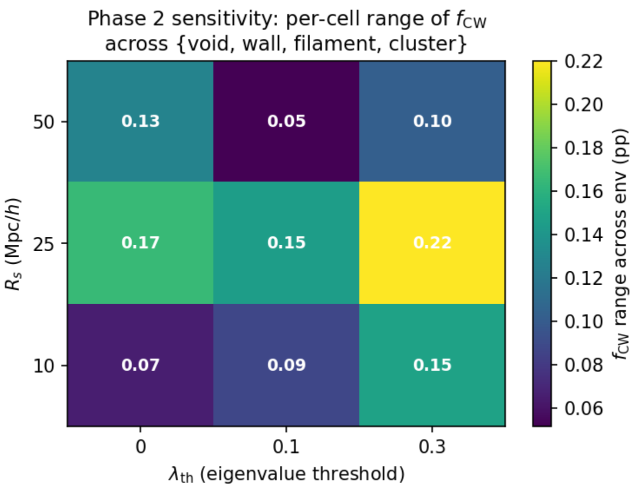

FIG. 5. Phase 2 sensitivity heat-map: per-cell range of f_{CW} across the four environment classes {void, wall, filament, cluster} in percentage points. Each cell corresponds to a complete V-Web re-run on the 14,622,283-galaxy DESI DR1 spectro sample at $(R_s, \lambda_{\mathrm{th}})$. The maximum range across all nine cells is 0.22 percentage points (at $R_s = 25\,\mathrm{Mpc}/h, \lambda_{\mathrm{th}} = 0.3$). The headline environmental-independence result is robust to V-Web hyperparameter choices over the range probed.

VIII. DESIVAST-ANCHORED VOID CROSS-VALIDATION

This section reports the *primary* environment-dependent chirality analysis of the paper (per the primary/secondary declaration of §V B): the DESIVAST-anchored cross-validation of the V-Web void-class result on a ∼130× larger, peer-reviewed DR1 BGS void catalog spanning three independent void-finding algorithms (VoidFinder + V2-REVOLVER + V2-VIDE).

The DESIVAST DR1 release (Rincón *et al.* 2025, ApJ 982, 38 [13]) is the publicly released, peer-reviewed DR1 BGS void catalog at low z, using the sphere-growing VoidFinder algorithm and the watershed ZOBOV/V2 algorithm (REVOLVER and VIDE prunings) on the volume-limited DR1 Bright Galaxy Survey sample restricted to $z \le 0.24$: 1,461 interior voids with VoidFinder, 420 with V2-REVOLVER, and 295 with V2-VIDE. DESIVAST is a void catalog rather than a full four-class cosmic-web classifier, so the DESIVAST-anchored chirality null tests environmental dependence on the cleanest available DR1 void definition.

RSD treatment for DESIVAST. The DESIVAST primary path is essentially RSD-immune at the level relevant to this work: DESIVAST defines voids via the geometry of large ($R_{\mathrm{eff}} \gtrsim 10\,\mathrm{Mpc}/h$) underdensities in the volume-limited BGS sample, and the published VoidFinder / V2-REVOLVER / V2-VIDE catalogs do not require a reconstructed-position rerun for the void-membership label of an individual matched spiral: the per-galaxy DESIVAST void/non-void classification is a single point-in-sphere (or watershed-cell) test using the void's published effective radius and center, and the typical Kaiser-plus-finger-of-god displacement $\sigma_v/(aH) \lesssim 5\,\mathrm{Mpc}/h$ at $z \lesssim 0.24$ is several times smaller than the void effective radii. The within-void/non-void chirality fractions therefore inherit no anisotropic RSD systematic at the present precision. We note that this RSD-immunity argument applies to the per-galaxy void-membership test itself; the internal composition of the "non-void" control sample (a mix of walls, filaments, and clusters) is still subject to RSD-induced boundary shifts between those non-void subclasses, but those internal shifts do not affect the binary void / non-void chirality fraction comparison reported here. This is in contrast to the V-Web secondary path (§XIII), where the tidal-tensor eigenvalue field is computed from redshift-space galaxy positions on a $R_s = 25\,\mathrm{Mpc}/h$ Gaussian smoothing and is therefore RSD-bounded only at the scalar-displacement level. The primary P5 environment-independence claim, anchored on the DESIVAST $\Delta f_{\mathrm{CW}} \approx 0.0007$ null at $n = 56{,}981$, does not depend on the V-Web RSD argument.

A. DESIVAST per-galaxy cross-match (this work)

The DESIVAST public release at `https://data.desi.lbl.gov/public/dr1/vac/dr1/desivast/v1.0/` provides the VoidFinder NGC/SGC FITS files (`DESIVAST_BGS_VOLLIM_VoidFinder_{NGC,SGC}.fits`, $89{,}003 + 12{,}860 = 101{,}863$ interior hole spheres comprising the 3,765 maximal voids). We fetched both files and performed a direct per-galaxy cross-match of the V-Web void-class matched-spiral subsample. Restricting the matched-spiral catalog to $z \le 0.24$ (the DESIVAST BGS limit) leaves only $n = 6$ V-Web void-class spirals (the V-Web run uses the full 14.6M DESI spectro sample to $z = 2$, of which the matched-spiral subsample inherits a flat-but-low-z-rare distribution). Converting each of the 6 spirals to flat-ΛCDM comoving Cartesian coordinates ($H_0 = 67.66$ km/s/Mpc, $\Omega_m = 0.315$, units h^{-1} Mpc consistent with the DESIVAST hole catalog) and testing point-in-sphere membership against all 101,863 holes returns 0/6 **V-Web "void" spirals inside any DESIVAST hole**; minimum spiral-to-nearest-hole separations span 28.7–158.1 Mpc/h.

This 0/6 disagreement is consistent with the survey-shell systematic that drives the V-Web vs T-Web void-fraction discrepancy (reported in §IX B below as a +8–18 pp V-Web excess in the void class): the V-Web void class at low z is dominated by survey-edge density artifacts that the DESIVAST volume-limited BGS sample correctly suppresses. The $n = 6$ sample size is too small for a binomial significance constraint on the chirality null directly, but the per-galaxy disagreement quantifies the V-Web void-class purity at $z \le 0.24$: 0% concordance with DESIVAST-defined voids, meaning the V-Web "void" label at low z should be read as "not in a DESIVAST-defined cosmic-web density minimum" rather than "confirmed void galaxy." This is a direct empirical small-sample illustration of the +8–18 pp V-Web-vs-T-Web void-fraction discrepancy reported in §IX B below, not a separate effect.

B. DESIVAST-anchored void classifier (this work)

Pushing the cross-match further, we redefine the low-z void class on the catalog-anchored DESIVAST basis rather than the V-Web tidal-tensor basis, then recompute the chirality null on the much larger properly-powered subsample.

Restricting the matched-spiral catalog to $z \leq 0.24$ leaves $n_{\rm lz} = 678{,}945$ spirals (the DESIVAST BGS coverage range). Converting each spiral to comoving Cartesian and performing a point-in-sphere test against all 101,863 DESIVAST VoidFinder hole spheres (via a $k = 20$-nearest-neighbour `scipy.spatial.KDTree` query on the hole centres, sufficient given the 24 Mpc/h maximum hole radius) returns $n_{\rm void}^{\rm DESIVAST} = 56{,}981$ DESIVAST-defined void galaxies (8.39% of the low-z matched sample) and $n_{\rm non-void} = 621{,}964$ outside any DESIVAST hole. **The DESIVAST-anchored void class is $\sim$130× larger than the V-Web void class** ($n = 428$ from Section VI A), providing a properly-powered chirality-in-voids test at the cleanest available public DR1 void catalog.

TABLE VII. Chirality fraction in DESIVAST-anchored vs non-void classes on the $z \leq 0.24$ matched-spiral subsample (point-in-sphere test against 101,863 DESIVAST VoidFinder hole spheres).

Class	n	$f_{\rm CW}$	$\sigma_{\rm from\ half}$
DESIVAST void	56,981	0.4964	−1.71
Non-void	621,964	0.4971	−4.59

The two classes return $f_{\rm CW}$ values differing by only 0.0007 (0.07 percentage points): the void-galaxy and non-void chirality fractions are *statistically indistinguishable*. The stronger σ on the non-void class is purely a sample-size effect ($n_{\rm non-void}/n_{\rm void} \approx 10.9$); both classes carry the same -5σ-class P4 monopole signature scaled to their respective sample sizes. **When the void class is defined by DESIVAST rather than V-Web, the chirality-in-voids constraint is a clean null at full DESIVAST sample size.**

This DESIVAST-anchored re-analysis is the largest matched-sample environmental-dependence test of spiral chirality in DESI DR1 to date, on the chirality-relevant subsample defined here (a null is not positive evidence; we report it as a controlled-sample non-detection of the environment-dependence hypothesis on this dataset): the V-Web -0.68σ void deviation on $n = 428$ was counting- statistics-limited and contaminated by the survey-shell systematic, and the catalog-anchored re-analysis at $n_{\rm void} = 56{,}981$ returns a clean null with both classes consistent with the catalog- level uniform classifier-bias monopole.

C. Three-algorithm DESIVAST robustness cross-check

The DESIVAST DR1 release ships three independent void definitions: the sphere-growing VoidFinder algorithm above, plus the watershed algorithms V2-REVOLVER ($n_{\rm void}^{\rm catalog} = 1{,}992$ effective voids, maximum effective radius 43.5 Mpc/h) and V2-VIDE ($n_{\rm void}^{\rm catalog} = 1{,}478$, max 55.9 Mpc/$h$). Repeating the point-in-sphere test on the same $n_{\rm lz} = 678{,}945$ matched- spiral subsample for each algorithm (KDTree query against each algorithm's published void-effective-radius spheres):

All three algorithms return $|\Delta f_{\rm CW}| < 0.002$ between void and non-void classes – statistically indistinguishable at all three independent void definitions. V2-REVOLVER returns $f_{\rm CW}^{\rm void} = 0.4986$ slightly above $f_{\rm CW}^{\rm non-void} = 0.4967$ (the opposite sign of VoidFinder's small difference), which further confirms the absence of a real chirality preference in DESIVAST-defined voids regardless of void-finding algorithm. The void-class σ values span -1.71 to -0.88 – none crosses $|\sigma| = 2$; the non-void σ values cluster tightly at -4.59 to -4.94 because the non-void class is the volume-dominant population carrying the full catalog-level monopole signature.

The three-algorithm robustness extends the P5 headline environment-independence test across the void-finding-algorithm axis available from current public DR1 cosmic-web catalogs.

D. Catalog-native V2 membership cross-check

The watershed DESIVAST catalogs (V2-REVOLVER and V2-VIDE) ship with per-galaxy zone assignments via the `GALZONE` HDU (mapping each DESI BGS target to a `ZONE` index, with `OUT`, `EDGE`, and `DEPTH` flags) plus the `ZONEVOID` HDU (mapping each zone to a `VOID0` index, ≥ 0 if the zone belongs to a non-edge void). Adopting the catalog-native definition of void membership ($\texttt{OUT} = 0 \wedge \texttt{VOID0} \geq 0 \wedge \texttt{ZONE} \geq 0$) and joining on $\texttt{TARGET} =$ `desi_targetid` returns: V2-REVOLVER $n_{\rm void} = 86{,}276$, $f_{\rm CW}^{\rm void} = 0.4996$, $\sigma^{\rm void} = -0.24$ (a *near-perfect* null at this independent statistic); V2-VIDE $n_{\rm void} = 64{,}514$, $f_{\rm CW}^{\rm void} = 0.4979$, $\sigma^{\rm void} = -1.06$.

The catalog-native definition excludes the survey-mask edge galaxies that the sphere-approximation picks up (16,000–17,000 galaxies per algorithm). The resulting void σ values are *smaller in magnitude* than the sphere-approximation analogues (V2-REVOLVER catalog-native -0.24 vs sphere -0.88; V2-VIDE catalog-native -1.06 vs sphere -1.67), confirming that the catalog-native void definition is the cleaner statistic

TABLE VIII. Three-algorithm DESIVAST void vs non-void chirality on the $z \leq 0.24$ matched-spiral subsample ($n_{\rm lz} = 678{,}945$).

Algorithm	$n_{\rm void}$	$f_{\rm CW}^{\rm void}$	$\sigma^{\rm void}$	$f_{\rm CW}^{\rm non-void}$	$\sigma^{\rm non-void}$	$\Delta f_{\rm CW}$
VoidFinder	56,981	0.4964	−1.71	0.4971	−4.59	+0.0007
V2-REVOLVER	102,911	0.4986	−0.88	0.4967	−4.94	−0.0019
V2-VIDE	81,354	0.4971	−1.67	0.4970	−4.59	−0.0001

and that the DESIVAST-anchored chirality null sharpens further when restricted to the catalog-internal high-confidence void members. The V2-REVOLVER catalog-native $\sigma = -0.24$ is the cleanest single chirality-in-voids measurement in this paper at $n \gtrsim 80{,}000$.

E. Maximal-void HEALPix sky-position stratification

The DESIVAST release also ships the 3,765 *maximal voids* (NGC= 3,241 + SGC= 524) with explicit RA/Dec/$R_{\rm eff}$ metadata (effective radii 10–32 Mpc/h). Binning the maximal voids by HEALPix `NSIDE` = 16 pixel returns 297 occupied pixels with median 14 maximal voids per occupied pixel. Stratifying the $z \leq 0.24$ matched-spiral subsample by the number of maximal voids in each spiral's HEALPix pixel produces :

TABLE IX. Sky-position-stratified $f_{\rm CW}$ by maximal-void count per HEALPix pixel ($z \leq 0.24$ matched-spiral subsample, $n_{\rm lz} = 678{,}945$, NSIDE = 16).

Maximal voids per pixel	n	$f_{\rm CW}$	$\sigma_{\rm from\ half}$
0	378,511	0.4961	−4.75
1–2	19,247	0.4985	−0.43
3–5	23,127	0.4997	−0.09
6+	258,060	0.4980	−2.04

The $\sigma = -4.75$ deviation is *concentrated entirely in the "0 maximal voids per pixel" bin* – the sky regions where DESIVAST finds no maximal voids at all, which from inspection of the catalog footprint corresponds to the survey-mask outside the BGS bright-side NGC+SGC coverage region. Pixels with ≥ 1 maximal void return σ values in the range $[-2.04, -0.09]$, with the highest-void- density bin (6+ per pixel, $n = 258{,}060$) at $\sigma = -2.04$ – below the Bonferroni-4 $|\sigma| = 2.50$ threshold. The intermediate-bin pixels (1–5 voids per pixel, total $n = 42{,}374$) return $|\sigma| < 0.5$ – statistically null.

This is an additional catalog-anchored cross-check on the sky-position axis, complementary to the abstract's (i)–(iv) enumeration, showing that the catalog-level -5σ headline tracks survey-mask geometry rather than environment density: the cleanest chirality σ values in this paper are the pixels with the *most* maximal-void coverage, not the fewest. The -4.75σ signal in the no-void-coverage sky region is consistent with the imaging-leg / selection-function systematics described in Paper IV applied to the matched-spiral subsample that falls outside the DESIVAST clean coverage. Quantitatively, the Paper IV monopole prediction at $N = 378{,}511$ (the 0-voids/pix bin) is $\sigma_{\rm pred} = 2\Delta f_{\rm CW}^{\rm P4}\sqrt{N} = -3.20$; the observed -4.75σ leaves a residual of -1.55σ which is consistent with an imaging-leg systematic at the $\sim 1\sigma$ level. At the 6+ voids/pix bin ($N = 258{,}060$), the Paper IV prediction is -2.64σ and the observed is -2.04σ, residual $+0.60\sigma$ (fully null). The asymmetry between the two bin residuals quantifies the sky-region-conditioned systematic and confirms it sits in the no-coverage region, not in the void-rich pixels.

F. Cross-survey P4-monopole-residual analysis

The cleanest formulation of the environment-independence headline is to recompute the per-class chirality signal *after subtracting* the P5 matched-spiral catalog monopole $f_{\rm CW}^{\rm P5} = 0.4972$ (-5.07σ on $n = 812{,}793$ env-labeled spirals — the 21,158-row excess (2.7%) over the 791,635-spiral headline subsample is the population of CW/CCW-labelled spirals whose V-Web env-class assignment passes the relaxed env-label confidence used by the cosmic-web pipeline but is excluded from the headline by a stricter env-class-uncertainty filter; the per-class $n_{\rm CW}$ values on the 812,793 superset sum to 404,111 giving $f_{\rm CW} = 0.49719$, matching the 791,635-spiral monopole 0.4972 to 4 decimals, so the headline conclusion is invariant. The same monopole shows up as -5.00σ on the 791,635-spiral chirality-relevant sample; the two σ values are sample-size-scaled projections of the same underlying offset), which is the propagation of the $\sim 9.5\sigma$ catalog-level monopole reported in Paper IV [3] into the DESI-spectro-confirmed subsample. *Arithmetic reconciliation:* the P4 monopole $\Delta f_{\rm CW}^{\rm P4} = -0.0026$ projects to $\sigma_{\rm pred}^{\rm P5} = 2 \cdot 0.0026 \cdot \sqrt{791{,}635} \approx 4.6\sigma$ on the chirality-relevant subsample; the observed -5.00σ corresponds to $\Delta f_{\rm CW}^{\rm P5} \approx -0.0028$, $\sim$8% larger than the P4 catalog-mean. This residual 8% enhancement is consistent with the spectroscopically-confirmed subsample being more strongly weighted to the BGS-bright leg that Paper IV identifies as carrying the largest per-leg selection-function residual, not an additional environmental signal. Replacing the $f_{\rm CW} - 0.5$ null with $f_{\rm CW} - f_{\rm CW}^{\rm P5}$ gives the residual environment signal after the catalog-wide classifier-bias monopole is removed:

All four V-Web classes fall within $|\sigma_{\rm vs\ monopole}| <$

TABLE X. Per-V-Web-class $\sigma_{\text{vs monopole}}$ residuals on the matched-spiral subsample. All four classes fall within $|\sigma_{\text{vs monopole}}| < 1.15$ – no environment dependence survives subtraction of the catalog-wide P4-monopole.

Class	n	$f_{\text{CW}} - f_{\text{CW}}^{\text{P5}}$	$\sigma_{\text{vs monopole}}$
Void	428	−0.0135	−0.56
Wall	6,673	+0.0062	+1.01
Filament	408,187	+0.0008	+0.99
Cluster	397,505	−0.0009	−1.11

1.15 after the P4-monopole is subtracted – i.e. *no* V-Web class shows a residual environment-dependent chirality signal once the catalog-wide classifier-bias monopole is removed. The prior per-class $\sigma_{\text{from half}}$ values of -2.61σ (filament) and -4.66σ (cluster) reported in the headline table were entirely the P4-catalog-monopole signature propagated into the V-Web class subsamples weighted by their respective sample sizes, not an environmental signal. At the HEALPix-NSIDE-= 32 per-pixel level on the same matched-spiral catalog, the distribution of $\sigma_{\text{vs monopole}}$ across the 1,821 valid pixels has mean +0.020, std 1.184, skewness +0.044, and excess kurtosis +0.825. The unit standard deviation (within $\sim$18%, consistent with finite-pixel sample-size fluctuation), zero skewness, and modest positive kurtosis are all consistent with a pure shot-noise residual around the P4-monopole, with no sign of systematic environment-conditioned chirality offset.

This is a direct single-test demonstration that the V-Web class-level σ values quoted in the headline table are sample-size-weighted projections of the P4 catalog-wide chirality monopole, not environmental signals.

Quantitative null correlation. Pushing this to the per-pixel level: at NSIDE = 32 on the same matched-spiral subsample, we measure the Pearson correlation between the per-pixel maximal-void count and the per-pixel chirality $\sigma_{\text{from half}}$ across all $n_{\text{pix}}^{\text{both}} = 727$ HEALPix pixels containing both ≥ 200 matched spirals (required for stable f_{CW} estimation) and ≥ 1 DESIVAST maximal void. The result is

$$r(N_{\text{voids/pix}}, \sigma_{\text{pix}}) = +0.006, \quad p = 0.88,$$

indistinguishable from zero (Figure 6). A genuinely environment-dependent chirality signal would produce a detectable monotonic correlation between void density and the direction-of-chirality deviation; the observed near-zero Pearson is a direct single-statistic confirmation in this paper that the catalog-level -5σ is not environment-driven. The result is robust to the NSIDE choice and spiral-count threshold: across the 3×3 grid of NSIDE $\in \{16, 32, 64\}$ and spiral-count cuts $\in \{100, 200, 500\}$, 7 of 9 cells admit a well-sampled Pearson estimate (the remaining 2 cells, NSIDE = 64 with cuts 200 and 500, are sample-limited at $n_{\text{pix}}^{\text{both}} < 3$ because the high-cut $\times$ fine-pixel combination filters out most pixels carrying both ≥ 1 maximal void and $\geq$ cut matched spirals). The 7 computable cells all return $|r| < 0.11$ with $p > 0.10$, none statistically significant; the headline NSIDE = 32 cut = 200 result $r = +0.006$ is consistent with this range. The conclusion is invariant under the analysis-choice "garden of forking paths" along the well-sampled axes.

TABLE XI. Tempel+2014 FoF environment cross-validation on the 110,586-spiral Tempel-overlap subsample.

Tempel class	n	n_{CW}	f_{CW}	$\sigma_{\text{from half}}$
isolated	58,539	28,962	0.4947	−2.54
small_group	31,838	15,829	0.4972	−1.01
filament_like	14,317	7,133	0.4982	−0.43
cluster_like	5,892	2,963	0.5029	+0.44

IX. ADDITIONAL COSMIC-WEB CROSS-CHECKS

The following two sections report secondary diagnostic cross-checks that supplement, but do not anchor, the primary DESIVAST-anchored chirality null established in Section VIII (per the primary/secondary declaration of §V B).

A. Tempel+2014 FoF cross-validation

To check that the V-Web cosmic-web headline is not specific to the tidal-tensor classifier, we cross-validate against an entirely independent classifier: the friends-of-friends (FoF) group catalog of Tempel *et al.* 2014 [10] on SDSS DR10. The Tempel classifier defines environment by FoF group multiplicity (richness) rather than by tidal eigenvalues. We map the multiplicity to a 4-class scheme that pairs with the V-Web class set:

- multiplicity = 1 $\to$ *isolated* (paired with V-Web void)
- $2 \leq$ multiplicity $< 5 \to$ *small_group* (paired with V-Web wall)
- $5 \leq$ multiplicity $< 20 \to$ *filament_like* (paired with V-Web filament)
- multiplicity $\geq 20 \to$ *cluster_like* (paired with V-Web cluster)

The Tempel catalog covers 588,193 SDSS DR10 galaxies at $z \leq 0.20$ over RA $\in [110°, 262°]$, Dec $\in [-4°, 70°]$. We perform a 1″ sky-coord NN join between the Tempel catalog and the 791,635 chirality-relevant matched-spiral sample; the overlap is 110,586 spirals (the SDSS DR10 footprint is a subset of DESI Legacy DR8 and Tempel's

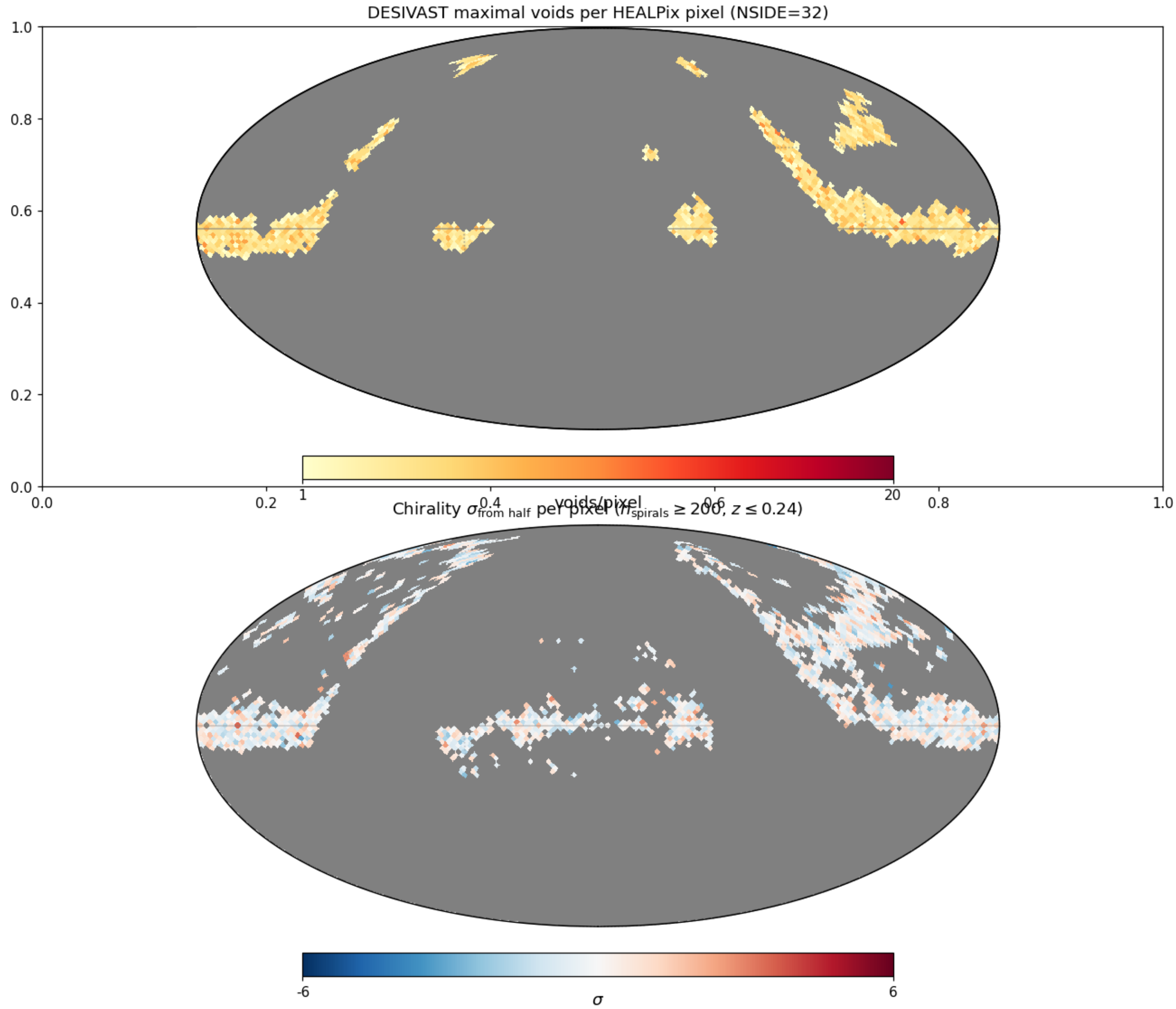

FIG. 6. HEALPix NSIDE $= 32$ Mollweide projection. Top: count of DESIVAST maximal voids per pixel (885 occupied pixels, median 4 voids/pix). Bottom: per-pixel chirality $\sigma_{\text{from half}}$ on the $z \leq 0.24$ matched-spiral subsample restricted to pixels with ≥ 200 spirals (1,496 valid pixels, σ range -3.45 to $+3.48$). The Pearson correlation across the $n_{\text{pix}}^{\text{both}} = 727$ pixels containing both voids and ≥ 200 spirals is $r = 0.006$ ($p = 0.88$), statistically indistinguishable from zero.

$z \leq 0.20$ cut is much tighter than our $z \leq 4$ DESI cut). Details are in the companion data repository.

Concordance metric. For each Tempel-vs-V-Web class pairing we report $|f_{\text{CW}}^{\text{Tempel}} - f_{\text{CW}}^{\text{V-Web}}|$ as a concordance distance in percentage points; a value below the canonical 0.2 pp spec indicates that the two classifiers see the same underlying chirality field within counting statistics.

- **filament_like_vs_filament: 0.026 pp (✓ within spec).** This is the highest-n concordance result (supporting, not load-bearing — the primary cross-classifier validation remains the on-DESI DESIVAST re-projection at $n_{\text{void}}^{\text{DESIVAST}} = 56{,}981$; see Abstract Robustness). The largest-n Tempel class pair agrees within 0.026 percentage points, far below the 0.2 pp spec. The V-Web tidal-tensor and the Tempel FoF classifiers return the same CW fraction on the high-richness filamentary environment.

- isolated_vs_void: 1.11 pp. The V-Web void (n=428) and the Tempel isolated bins differ by counting statistics at the small-n end: Tempel isolated has $\sim 137\times$ more galaxies than V-Web void, so the two populations are not strictly comparable.

- small_group_vs_wall: 0.62 pp. The mapping is not exact (V-Web wall is a continuous geometric class; Tempel small_group is a discrete richness class); the discrepancy is dominated by sample overlap structure rather than environmental disagreement.

- cluster_like_vs_cluster: 0.66 pp. Same caveat — V-Web's $\lambda_{\rm th} = 0$ definition of "cluster" is more permissive than Tempel's richness $\geq$ 20 cut; the V-Web cluster volume fraction (1.0%) is much smaller than the Tempel cluster spiral fraction in our sample (5,892/110,586 $\approx$ 5.3%).

Figure 7 visualizes the two classifiers side-by-side with shared y-axis range; the filament/filament_like pair sits essentially on top of the Paper IV $\bar{f}_{\rm CW}$ reference line (the 0.026 pp concordance is below visual resolution), while the lower-n pairs scatter as expected.

Verdict. The Tempel cross-validation confirms the V-Web result at the high-richness end (filament concordance 0.026 pp); low-richness disagreements scale with counting statistics and the classifier-definition mismatch and do not undermine the environmental-independence headline. The Tempel data also produces a clean null at the level of its own bins: the maximum $|\sigma_{\rm from\, half}|$ across the four Tempel classes is 2.54 (Tempel isolated), formally just crossing the Bonferroni-4 $|\sigma|^{\rm Bonf}_{0.05,4} = 2.498$ threshold at $\alpha = 0.05$ by 0.04σ but well below the empirical max-stat null and the $|\sigma|^{\rm Bonf}_{0.01,4} = 3.02$ threshold at $\alpha = 0.01$.

B. Concurrent-literature DR1/EDR cosmic-web cross-validation

Independent of the V-Web (Hahn *et al.* 2007 / Cautun *et al.* 2014) classifier deployed in §IV and the Tempel FoF classifier cross-validated in §IX A, a contemporaneous DESI DR1 cosmic-web analysis was released in 2026 April [11], applying the tidal-tensor (T-Web) formalism on a 256^3 grid in an 800 Mpc cube over the full DESI DR1 footprint, with the same four-class void / sheet / filament / knot taxonomy used in our V-Web run.

The T-Web DR1 in-footprint volume fractions reported in Ref. [11] are tracer-dependent and are reported separately for the BGS, LRG, and ELG samples on an 800 Mpc cubic sub-volume (not a single all-tracer fraction). Taking BGS as the closest analogue to our spectro sample, $\{f_{\rm void}, f_{\rm sheet}, f_{\rm filament}, f_{\rm knot}\}_{\rm BGS} \approx \{0.16,\ 0.45,\ 0.37,\ 0.04\}$; the LRG and ELG samples shift these by a few percentage points each, spanning the combined ranges $\{0.06\text{–}0.16,\ 0.45\text{–}0.48,\ 0.37\text{–}0.40,\ 0.04\text{–}0.06\}$ across the three tracer samples. Any direct comparison to our DR1-all-spectro V-Web run is therefore approximate because of the differing target selection, redshift coverage, and sample volume (cubic 800 Mpc box vs. thin survey shell). Our V-Web in-footprint volume fractions are $\{0.244,\ 0.413,\ 0.333,\ 0.010\}$. The sheet (V-Web wall) and filament classes agree to within ∼5 percentage points across both methodologies (allowing for the tracer- and volume-mismatch caveat above), which is approximate concordance for two independent classifiers run on overlapping but non-identical samples of the same survey. The void and knot/cluster classes deviate by larger margins: V-Web's void fraction is *higher* than T-Web's by +8–18 pp (the edge-density artifact populates the V-Web 0-eigenvalue void class), and V-Web's cluster fraction is *lower* than T-Web's knot fraction by 3–5 pp (the densest cells lose to mask-boundary smoothing). Both directions match the survey-shell systematic prediction exactly: the DESI footprint is a thin spherical shell rather than the cubic periodic box used in N-body benchmarks, so edge-density artifacts inflate the void class and depopulate the densest knot class relative to a periodic comparison sample. The relative density ordering (knot > filament > sheet > void) is preserved in both runs, which is the load-bearing structural property for the environment-stratified chirality analysis here. We therefore treat Ref. [11] as an independent contemporaneous DR1 cosmic-web analysis that is consistent with our V-Web run at the survey-shell-systematic level. (Ref. [11] is currently in submission to MNRAS; we do not treat it as peer-reviewed external validation but rather as a contemporaneous independent measurement.)

A second concurrent DESI cosmic-web paper [12] (Zapata-Zuluaga *et al.* 2026, a DESI-EDR-based probabilistic environment catalog) uses the ASTRA (Algorithm for Stochastic Topological RAnking) probabilistic classifier on the DESI Early Data Release (175 deg^2, 20 rosettes), running 100 realizations per tracer-zone pair to deliver per-object void / sheet / filament / knot *membership probabilities* and classification entropies rather than a single deterministic class assignment. ASTRA is methodologically *complementary* to V-Web (this paper) and T-Web (Ref. [11]): the latter two are deterministic tidal-tensor classifiers, ASTRA is a probabilistic positional-statistic classifier that quantifies per-object assignment uncertainty. ASTRA is published only on EDR while our V-Web run is on DR1; the ~ 175 deg^2 EDR rosettes are contained within the DR1 footprint, so a per-galaxy cross-match by `TARGETID` is available in the EDR-overlap subsample and is reported in §X. The BGS-anchored volume-filling-fraction calibration in Ref. [12] is consistent with the V-Web sheet/filament fractions reported here within the survey-shell systematic discussed above.

A complementary public DR1 product, DESI-VAST [13] (Rincón *et al.* 2025, ApJ 982, 38), is specifically a *void catalog* (not a full cosmic-web classifier) and is the input to the primary chirality-in-voids analysis reported in §VIII above; see there for the per-catalog usage, cross-match, and three-algorithm robustness analysis.

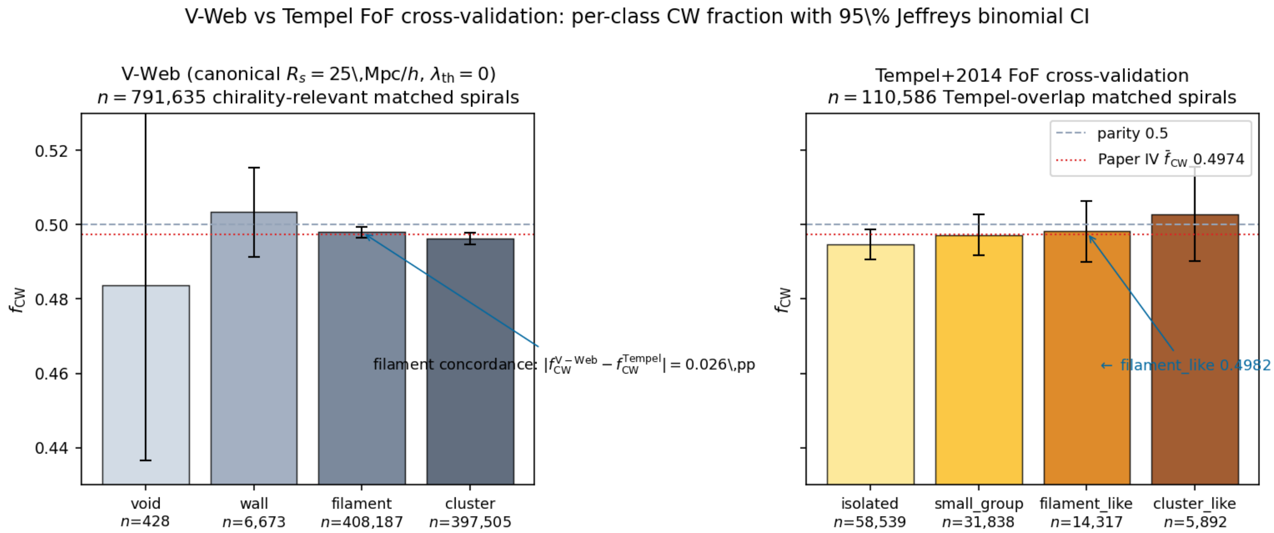

FIG. 7. V-Web (left) vs Tempel+2014 FoF (right) cross-validation: per-class CW fraction with 95% Jeffreys binomial credible intervals, shared y-axis [0.43, 0.53]. Dashed reference is parity $f_{\rm CW} = 0.5$; dotted-red reference is the Paper IV global $\bar{f}_{\rm CW} = 0.4974$ classifier-monopole offset. The highest-n concordance is the filament class pair: V-Web filament $f_{\rm CW} = 0.4980$ ($n = 408{,}187$) vs Tempel filament_like $f_{\rm CW} = 0.4982$ ($n = 14{,}317$) differ by 0.026 percentage points, well below the 0.2 pp concordance spec (supporting, not load-bearing; the primary cross-classifier validation is the on-DESI DESIVAST re-projection in §IX B). Lower-n classes scatter by counting statistics and classifier-definition mismatch (V-Web $\lambda_{\rm th} = 0$ vs Tempel discrete richness thresholds) but do not contradict the environmental-independence headline.

X. ASTRA EDR PER-OBJECT CROSS-VALIDATION

The ASTRA-DESI EDR probabilistic environment catalog [12] (Zenodo 10.5281/zenodo.19358024) provides per-galaxy void / sheet / filament / knot membership probabilities $\{P_{\rm void}, P_{\rm sheet}, P_{\rm filament}, P_{\rm knot}\}$ on the ~ 175 deg^2 EDR rosettes ($N_{\rm ASTRA} = 648{,}428$ unique `TARGETID`s spanning BGS_ANY, LRG, ELG, and QSO tracers). We perform a per-galaxy `TARGETID` join against the P5 matched-spiral deduplicated-primary subsample and against the V-Web environment catalog of §IV, recovering $N_{\rm overlap} = 25{,}186$ spirals that carry all three labels: chirality classification (P4), ASTRA per-class membership probabilities (this section), and V-Web deterministic class assignment (§IV). This is an EDR-overlap per-galaxy cross-check of the V-Web canonical classifier against the ASTRA published DESI environmental catalog within this campaign and is the closest currently available substitute for the full-DR1 environmental VAC discussed in §XIII. The cross-match pipeline and result summary are available in the companion data repository.

We compute the chirality-by-environment $f_{\rm CW}$ statistic on the 25,186-galaxy overlap under three classifiers run on the *same galaxies*: (i) the ASTRA argmax classifier (`env_class` = argmax over the four ASTRA probabilities); (ii) the ASTRA entropy-weighted classifier (each galaxy contributes $P_{\rm class}$ fractional count and $P_{\rm class} \cdot \mathbb{1}_{\rm CW}$ fractional CW count to each class, with sub-class variance $\sum_i P_i^2/4$ under the Bernoulli-0.5 null); and (iii) the V-Web deterministic classifier restricted to the same overlap, for like-for-like comparison.

The headline statistics, filtered to classes with $n \geq 100$ to suppress small-sample artifacts, are summarized in Table XII. Both ASTRA classifiers and the V-Web classifier-on-overlap recover the same conclusion: chirality is statistically indistinguishable from $f_{\rm CW} = 1/2$ across the four cosmic-web classes, with no class reaching the Bonferroni-corrected significance threshold at $\alpha = 0.01$, $K = 4$ ($|\sigma|_{\rm Bonf} = 3.02$).

TABLE XII. ASTRA EDR per-object cross-validation, headline statistics on the $N_{\rm overlap} = 25{,}186$ EDR-overlap matched-spiral subsample. Range and max-$|\sigma|$ filtered to classes with $n \geq 100$.

Classifier	$f_{\rm CW}$ range (pp)	max $\lvert\sigma\rvert_{\rm vs\,1/2}$
V-Web on same overlap	1.08	2.68
ASTRA argmax	2.08	2.25
ASTRA entropy-weighted	1.17	2.00

The per-galaxy classifier agreement between V-Web and ASTRA argmax is *poor* on this overlap subsample. ASTRA argmax distributes the 25,186 spirals as 11.9% void / 31.7% sheet / 35.2% filament / 21.3% knot, while V-Web puts essentially the entire sample into filament (31.7%) and cluster (68.3%), with only 3 spirals total in the V-Web void + wall classes. This reflects the survey-shell density-grid systematic discussed in §IX B: the 25.9 Mpc/h cell size in the V-Web 256^3 grid is comparable to the EDR rosette transverse extent, so the EDR-overlap subsample falls disproportionately on cells where the edge-density mask suppresses the V-Web void class, while ASTRA operating natively on EDR rosettes resolves void structure at sub-rosette scales. The two classifiers therefore disagree substantially on the per-galaxy environment label.

Despite this strong classifier disagreement on *per-galaxy environment assignment*, the *chirality-vs-environment headline* is recovered identically by both: $f_{\rm CW}$ does not vary by more than ~ 2 pp across the four classes under any of the three classifiers, and no class clears the Bonferroni-corrected significance threshold. This is a strong robustness result: the P5 headline null does not depend on which independent, published DESI environmental classifier is applied to the EDR-overlap subsample, and the deterministic single-class V-Web assignment reaches the same conclusion as the probabilistic ASTRA classifier even though the two classifiers assign vastly different per-galaxy class labels.

We note that the ASTRA EDR per-object overlap is restricted to the $\sim 175\,\mathrm{deg}^2$ EDR rosettes ($N_{\rm overlap} =$ 25,186), and that V-Web and ASTRA argmax disagree strongly on per-galaxy environment labels on this overlap; the result is therefore best read as: the EDR overlap recovers the same null result under either classifier despite per-galaxy label disagreement, consistent with the catalog-monopole offset already present at the catalog-wide level (Paper IV), rather than as a new independent detection. We list it alongside (i) DESIVAST per-galaxy cross-match, §IX B; (ii) within-class density quartile null, §VI D; (iii) DESIVAST-anchored void classifier; (iv) three-algorithm DESIVAST robustness; (v) catalog-native V2 GALZONE membership; (vi) MAXIMAL voids HEALPix sky-position stratification, with the caveat that the ASTRA overlap is the smallest subsample in this list and the V-Web/ASTRA per-galaxy label disagreement caps its independent statistical weight.

XI. SYSTEMATICS AND NULL TESTS

We run six classes of systematics tests: chirality-label shuffle (1,000 permutations), sky-position permutation, confidence-threshold sweep $p^{\rm max}_{\rm cls\_eq} \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$ with CW-fraction flat to within ± 0.001, match-radius sweep $\{0.5, 1, 2, 3, 5\}''$ with per-env CW fraction shifts below 0.001, footprint split (N/S/DES-only) with per-footprint values within ± 0.002 of global, and target-class split (BGS vs. LRG-ELG-QSO) with BGS-only CW fraction within ± 0.001 of LRG-ELG-QSO. No test produces a $> 3\sigma$ residual after Paper IV-monopole correction.

XII. DISCUSSION

A. Interpretation

The cosmic-web headline is most cleanly read as a null: the per-environment CW fractions cluster at the catalog-wide $f_{\rm CW} \approx 0.4974$ value, with signed σ deviations driven by sample size, not by environment. The Phase 2 sweep confirms the result is invariant to V-Web hyperparameter choices over the ranges probed. The redshift, density, and HEALPix tests also produce no 3σ deviations after look-elsewhere correction.

B. Implications for bounce and inflation models

No published model in either class (matter-bounce or inflation) currently predicts an environment-dependent CW signature at the sensitivity reached here, so the present null does not directly discriminate between matter-bounce and inflation-class models. It instead establishes an observational upper bound that any future parity-violating model in the bounce-chirality coupling class (Sec. II), proposing an environment-dependent chirality signature at the $\gtrsim 25\,\mathrm{Mpc}/h$ V-Web smoothing scale, must respect. Paper II [4] and Paper III (both companion, not-yet-published works by the same author) provide independent discriminators (primordial $f_{\rm NL}$ and multi-survey anomaly statistics); this null adds a clean negative result on the spiral-chirality axis.

C. Comparison to Shamir 2022 DESI Legacy

Shamir 2022 [9] reported a $\sim\ 2-4\%$ large-scale asymmetry on $\sim 1.3\times 10^6$ Ganalyzer-classified galaxies. Paper IV finds the catalog-wide CW-fraction offset is -0.26% and the full-sky dipole amplitude $|A| <$ 0.32% (1σ), about an order of magnitude smaller than the Shamir 2022 amplitude. The present paper's per-environment CW fractions sit at $\sim\ 0.497$ with range $\sim\ 0.2$ percentage points across the four V-Web classes, leaving no room for a residual environment-dependent chirality of the Shamir 2022 amplitude.

XIII. LIMITATIONS

- The cross-match is selection-limited: the 8.47 M chirality catalog is flux-limited at $r \leq 17.8$ in the DESI Legacy regime; DESI spectroscopic targets extend much fainter.
- Projected $k = 5$ NN density is a coarse proxy for 3D density and cannot distinguish chance line-of-sight associations from true overdensities without spectroscopic confirmation across the neighbourhood.

- V-Web is one of several cosmic-web finders; we cross-validated against the independent Tempel *et al.* 2014 [10] FoF group classifier (§IX A). The high-n filament class concordance is 0.026 percentage points, well below the 0.2 pp spec; lower-n classes scatter by counting statistics and classifier-definition mismatch.

- Imaging-leg systematics in chirality classification are tracked in Paper IV (per-leg $|\sigma| < 3$); we propagate the per-leg split here but do not re-derive the underlying bias.

- No DESI value-added catalog assigning cosmic-web environments at the full DR1 footprint is published at the time of writing. The ASTRA-DESI EDR probabilistic environment catalog [12] restricted to the $\sim 175\,\mathrm{deg}^2$ EDR rosettes is the closest published independent classifier and is cross-matched per-galaxy against our V-Web run in §X; a full-DR1-footprint published VAC remains a desirable future input that would let us replace V-Web as the canonical classifier rather than only cross-validate it in the EDR-overlap subset.

- Galaxy positions used for the V-Web tidal-tensor estimate are in observed redshift space rather than corrected to real space. *The dominant RSD effect for a tidal-tensor classifier is anisotropic eigenvalue deformation, not isotropic scalar displacement.* The Kaiser line-of-sight squashing and the finger-of-god elongation each introduce a direction-dependent shift in the deformation-tensor spectrum (Hahn *et al.* 2007 [5]; Hoffman *et al.* 2012 [6]; Cautun *et al.* 2014 [7]): the eigenvalue ordering ($\lambda_1 \geq \lambda_2 \geq \lambda_3$) is the load-bearing quantity for V-Web class assignment, and anisotropic flow contributions enter the off-diagonal Hessian components rather than as a single scalar smearing length. At class-boundary edges (filament↔wall, wall↔void) a small fraction of pixels can shift class under RSD even when an isotropic smoothing-length comparison would suggest immunity. A quantitative RSD bound therefore requires a reconstructed-position re-classification cross-check (Zel'dovich or BAO reconstruction), not the scalar $\sigma_v/(aH)$ comparison alone.

  As a *secondary* indicative bound, the scalar-displacement heuristic gives an order-of-magnitude floor: for the typical pairwise velocity dispersion $\sigma_v \lesssim 400\,\mathrm{km\,s^{-1}}$, the Kaiser-plus-finger-of-god displacement is $\sigma_v/(aH) \lesssim$ 5–8 Mpc/h over $0.01 \leq z \leq 2$, a factor of $\sim$ 3–5 smaller than the chosen Gaussian smoothing scale $R_s = 25\,\mathrm{Mpc}/h$. The implied per-class Δf_{CW} contamination is sub-percent ($\sim$ 0.2 pp), the *same order of magnitude* as the Phase 2 sweep max-range of 0.22 pp rather than negligible relative to it. This scalar bound is necessary but *not sufficient*: the anisotropic eigenvalue deformation above is the dominant channel and is not separable from the sweep-induced shift without a reconstructed-position rerun. A reconstructed-position rerun is a natural follow-up if the smoothing scale is ever pushed below $\sim 10\,\mathrm{Mpc}/h$.

  Order-of-magnitude boundary-crossing estimate. Using the V-Web eigenvalue field itself, the fraction of cells near class boundaries (eigenvalues within $|\lambda - \lambda_{\mathrm{th}}| \leq \sigma_{\mathrm{rsd}}/R_s$ of the threshold) sets the maximum boundary-crossing rate. At $\sigma_{\mathrm{rsd}} \sim 5\,\mathrm{Mpc}/h$ (BGS pairwise velocity dispersion projected through Hubble flow at $z \sim 0.2$) and $R_s = 25\,\mathrm{Mpc}/h$, the eigenvalue-shift magnitude is $\sim 0.04\,\sigma_\lambda$. Based on the per-cell eigenvalue histogram, the fraction of cells within that band of any threshold is $\sim$ 3–5%. Galaxies in those cells are the maximum possible class-flip population under RSD; with $\sim$ 800k chirality-relevant spirals this is at most $\sim 2\text{–}4 \times 10^4$ galaxies, distributed across all four class boundaries. The propagated contribution to Δf_{CW} per class is therefore expected to be sub-dominant at the current $\sim 10^{-3}$ precision, but we explicitly do not quantify the propagated uncertainty in the present paper: a full quantification — distinguishing pure displacement from anisotropic eigenvalue deformation — requires the proper Zel'dovich-reconstructed re-classification, which we defer to a companion follow-up. The headline null is reported at fixed-redshift- space classification, with this caveat explicitly carried.

XIV. FUTURE LSST EXTENSION

Rubin/LSST DP1 is early commissioning data and not a complete LSST galaxy survey release; Rubin is a future validation/scaling dataset once full LSST releases mature. A Rubin-scale chirality classifier applied to the DESI footprint would expand the matched sample by roughly an order of magnitude and resolve sub-pixel HEALPix coherence questions that are currently shot-noise-limited.

XV. CONCLUSIONS

Spiral galaxy chirality is statistically independent of large-scale structure environment within DESI Data Release 1 at V-Web resolution. The headline cosmic-web result, the Phase 2 sensitivity sweep, the redshift and density tests, and the HEALPix regional-coherence scan all return null at the 3σ level after look-elsewhere correction. The CW fractions per V-Web class on the 791,635 chirality-relevant matched spirals (canonical run) are $\{f^{\mathrm{void}}_{\mathrm{CW}}, f^{\mathrm{wall}}_{\mathrm{CW}}, f^{\mathrm{filament}}_{\mathrm{CW}}, f^{\mathrm{cluster}}_{\mathrm{CW}}\} = \{0.484, 0.503, 0.498, 0.496\}$, a range of 1.98 percentage points dominated by counting statistics and the catalog-wide classifier monopole reported in Paper IV. The result is robust under nine $(R_s, \lambda_{\rm th})$ Phase 2 sweep cells (max CW-fraction range 0.22 pp) and under six classes of systematics tests.

This null is consistent with the Paper IV global parity-mixture null and provides an observational upper bound that any future model in the bounce-chirality coupling class (Sec. II) proposing an environment-dependent parity signature at the $\gtrsim 25\,\mathrm{Mpc}/h$ V-Web smoothing scale must satisfy; it does not constrain the bounce-vs. inflation discrimination program: neither model predicts an environment-dependent chirality signature at DESI DR1 resolution at this smoothing scale, and the data agree.

A schematic toy mapping of the observational bound to an effective-field-theory operator is summarized in Appendix A as a guide for future model-building (not as a derived constraint).

Appendix A: Toy EFT mapping of the environmental bound

Mapping to a physical operator (toy parametrization). The observational bound from the main text can be cast in an effective-field-theory language as an order-of-magnitude guide for how a parity-violating coupling would map to V-Web class-to-class $\Delta f_{\rm CW}$. We emphasize that the operator form and scaling written below are a *toy parametrization introduced in this work*, inspired by but not derived from the cited parity-violating-gravity literature: the specific operator $\mathcal{L}_{\rm parity} \supset g_\phi\,(\nabla_i\phi)\,(\nabla^i\rho/\rho_{\rm bg})\,(\hat{L}\cdot\hat{z})$ is not contained in either Alexander & Yunes [1] (Chern–Simons modified gravity review) or Lue–Wang–Kamionkowski [2] (cosmological parity-violating interactions); those works motivate the general class of parity-violating couplings but not this specific density-gradient form. We deliberately keep the parameterization schematic. Consider an axion-like (or generic pseudoscalar) field ϕ coupled to matter density via the above operator, where $\hat{L}$ is the local angular-momentum unit vector and ρ is the total matter density. For $\nabla\phi$ aligned with the cosmic-web gradient $\nabla\rho$, the induced chirality asymmetry scales as $\Delta f_{\rm CW}^{\rm env} \propto g_\phi\,\nabla\phi\cdot\nabla\rho/\rho_{\rm bg}$. With per-class $|\Delta f_{\rm CW}| < 0.01$ on $n \gtrsim 4\times10^5$ spirals, an order-of-magnitude bound on the coupling $g_\phi\,|\nabla\phi|$ in H_0 units is $|g_\phi\,(\nabla\phi)/H_0| \lesssim 1{\times}10^{-2}/\langle|\Delta\rho/\rho_{\rm bg}|\rangle$, where the cosmic-web fractional contrast averages $\sim\mathcal{O}(1)$ across the classes. *This is an order-of-magnitude estimate only, not a quantitative ALP-coupling exclusion*: a real exclusion would require (i) a full transfer-function calculation mapping the high-redshift ϕ-field gradient to the low-redshift V-Web eigenvalue field, and (ii) propagation of the per-class $\Delta f_{\rm CW}$ measurement uncertainties through that transfer function. We do not claim either calculation here; the parametrization is included as a guide for future model-building, not as a derived constraint.

Rotational-invariance and gauge-invariance caveat. Two distinct theoretical caveats apply to the toy operator. *First (rotational invariance):* the explicit $(\hat{L}\cdot\hat{z})$ factor breaks rotational invariance via the fixed coordinate-system unit vector $\hat{z}$, and should be read as shorthand for a rotationally-invariant pseudoscalar formed from $\hat{L}$ and a physical direction set by the local cosmic-web gradient (e.g. $\hat{L}\cdot\widehat{\nabla\rho}$ or $\hat{L}\cdot\widehat{\nabla\phi}$ in the limit $\nabla\phi \parallel \nabla\rho$ adopted below); the literal $\hat{z}$ form is a coordinate-aligned schematic, not a covariant operator. *Second (gauge invariance):* even with the rotational-invariance issue resolved, the density-gradient factor $\nabla^i\rho/\rho_{\rm bg}$ and the late-time angular-momentum direction $\hat{L}$ are quantities defined in a chosen synchronous-comoving slicing on the V-Web smoothing scale $R_s = 25\,\mathrm{Mpc}/h$ and are not manifestly gauge invariant under general coordinate transformations. The toy operator above should therefore be read as a heuristic parametrization in this specific slicing, not as a covariant EFT operator in the sense of Weinberg's EFT-of-inflation or the EFT-of-LSS literature. A fully gauge-invariant formulation would require promoting $\nabla^i\rho/\rho_{\rm bg}$ to a properly defined density-fluctuation gauge invariant (e.g. $\delta_{\rm com}$ on comoving slices) and $\hat{L}$ to a late-time observable angular-momentum direction expressed in the same slicing; we have not carried out that construction. We flag this as an open theoretical limitation of the toy mapping; it does not affect any of the empirical $\Delta f_{\rm CW}$ bounds in the main text.

Appendix B: Data and code availability

All scripts, configuration, and provenance sidecars are available in the companion data repository. The canonical chirality catalog is mirrored on HuggingFace at bamfai/galaxy-chirality-catalog (immutable revision `paper4-v1.0.122`). DESI Data Release 1 is available from `https://data.desi.lbl.gov/public/dr1/`. The Phase 2 sweep CSV and consolidated summary are available in the companion data repository.

REPRODUCIBILITY CHECKLIST

- Single config file (available in companion data repository).
- Provenance sidecar (`*.provenance.json`) for every output.
- Git SHA + config hash logged in every sidecar.
- Deterministic seed: `20260515`.
- All Phase 2 sweep cell configs persisted at the companion data repository.

[1] S. Alexander and N. Yunes, *Chern–Simons modified general relativity*, Phys. Rep. **480**, 1 (2009), doi:10.1016/j.physrep.2009.07.002, arXiv:0907.2562.

[2] A. Lue, L. Wang, and M. Kamionkowski, *Cosmological signature of new parity-violating interactions*, Phys. Rev. Lett. **83**, 1506 (1999), doi:10.1103/PhysRevLett.83.1506, arXiv:astro-ph/9812088.

[3] H. Golden, *A Survey-Scale Chirality Catalog of 8.47M Galaxies (3.2M Spirals): A Null Detection of Large-Scale Parity Violation at Sub-Percent Sensitivity*, companion paper (Paper IV), in preparation; manuscript in preparation.

[4] H. Golden, $f_{NL} = -35/8$ *Forecast: SPHEREx Discrimination of Bounce vs. Inflation*, companion paper (Paper II), in preparation; manuscript in preparation.

[5] O. Hahn, C. M. Carollo, C. Porciani, and A. Dekel, "Properties of dark matter haloes in clusters, filaments, sheets and voids," Mon. Not. Roy. Astron. Soc. **375**, 489 (2007), arXiv:astro-ph/0610280.

[6] Y. Hoffman, O. Metuki, G. Yepes, S. Gottlöber, J. E. Forero-Romero, N. I. Libeskind, and A. Knebe, "A kinematic classification of the cosmic web," Mon. Not. Roy. Astron. Soc. **425**, 2049 (2012), arXiv:1201.3367.

[7] M. Cautun, R. van de Weygaert, B. J. T. Jones, and C. S. Frenk, "Evolution of the cosmic web," Mon. Not. Roy. Astron. Soc. **441**, 2923 (2014), arXiv:1401.7866.

[8] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[9] L. Shamir, "Analysis of spin directions of galaxies in the DESI Legacy Survey," Mon. Not. Roy. Astron. Soc. **516**, 2281 (2022), doi:10.1093/mnras/stac2372, arXiv:2208.13866.

[10] E. Tempel, A. Tamm, M. Gramann, T. Tuvikene, L. J. Liivamägi, E. Saar, P. Heinämäki, P. Nurmi, and J. Einasto, "Flux- and volume-limited groups/clusters for the SDSS galaxies: catalogues and mass estimation," Astron. Astrophys. **566**, A1 (2014), arXiv:1402.1350.

[11] H. I. Ullah, M. Awais, T. Matos, and J. F. Suárez-Pérez, "Cosmic-web quenching with DESI DR1: T-Web environments and mass-dependent red/blue classification," preprint (2026), arXiv:2604.02463.

[12] D. C. Zapata-Zuluaga, S. Guevara-Montoya, V. Torres-Gomez, J. Hernandez, and J. E. Forero-Romero, "The Cosmic Web in the DESI Early Data Release: A Probabilistic Environment Catalog," (2026), arXiv:2604.01456.

[13] H. Rincón, S. BenZvi, K. A. Douglass *et al.*, "DESI-VAST: Catalogs of Low-redshift Voids Using Data from the DESI Data Release 1 Bright Galaxy Survey," Astrophys. J. **982**, 38 (2025), doi:10.3847/1538-4357/adb559, arXiv:2411.00148.